

Graph Theory – Lecture Notes

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1 Introduction

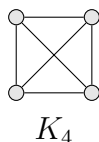
Definition 1.1. A (*simple*) *graph* G is a pair (V, E) where V is a non-empty set and $E \subseteq \{\{x, y\} \in V \times V : x \neq y\}$. We call the elements of V *vertices* of G and the elements of E *edges* of G . We also write $V(G)$ for V and $E(G)$ for E , respectively.

‘Simple’ means no loops and no double edges.

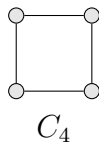
Remark 1.2. V is finite, unless otherwise stated.

Example 1.3. We define a few common graphs. Let $n, m \in \mathbb{N}$.

- The *complete graph* K_n has n vertices and each pair of vertices is connected.



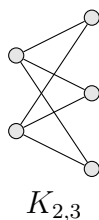
- The *cycle* C_n has n vertices, which are cyclically connected.



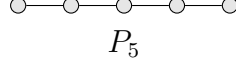
- Let A and B be disjoint sets with $|A| = n$ and $|B| = m$. The *complete bipartite graph* $K_{m,n}$ is defined by

$$V = A \dot{\cup} B \tag{1}$$

$$E = \{\{a, b\} : a \in A, b \in B\} \tag{2}$$



- The *path* P_n has the vertices $1, \dots, n$ and edges $\{a, a+1\}$ for $a \in \{1, \dots, n-1\}$.
(in books sometimes P_{n-1} has n vertices)



Definition 1.4. Let G be a graph, $u, v \in V(G)$.

- If $\{u, v\} \in E(G)$, then u and v are **adjacent** and the edge $e = \{u, v\}$ is **incident** with both u and v .
- $N_G(v) = \{w \in V(G) : \{v, w\} \in E(G)\}$ is the **neighbourhood** of v in G .
- $\deg_G(v) = |N_G(v)|$ is the **degree** of v .

Lemma 1.5 (Handshake Lemma). *Let G be a graph. Then*

$$\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|. \quad (3)$$

(in particular, the sum of degrees is even).

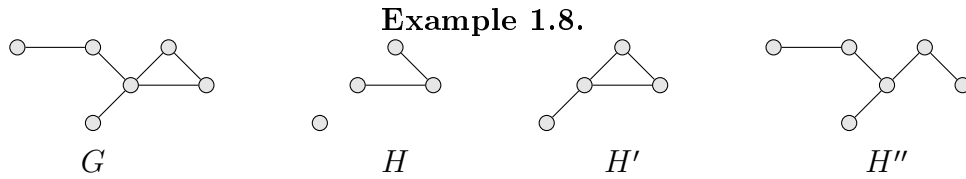
Proof. $\sum_{v \in V(G)} \deg_G(v) = |\{(v, e) : v \in V(G), e \in E(G) \text{ such that } v \text{ incident to } e\}| = 2|E(G)|$ $\square$

Definition 1.6. Two graphs $G = (V, E)$ and $G' = (V', E')$ are **isomorphic** if there exists an isomorphism ϕ between them, i.e., a bijection $\phi : V \rightarrow V'$ such that

$$\forall u, v \in V(G) : \{u, v\} \in E(G) \iff \{\phi(u), \phi(v)\} \in E(G').$$

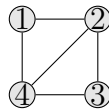
Definition 1.7. Let $G = (V, E)$ be a graph. A **subgraph** of G is a graph $H = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E \cap \binom{V'}{2}$, where $\binom{S}{K}$ denotes the set of all K -subsets of S .

- H' is an **induced subgraph** of G if it is a subgraph with $E' = E \cap \binom{V'}{2}$.
- H'' is a **spanning subgraph** if it is a subgraph with $V' = V$.
- For $W \subseteq V$, we write $G[W]$ for the induced subgraph on W .



Definition 1.9. Let v_i be vertices of a graph G .

- A **walk** is a sequence of vertices v_i s.t. $\{v_i, v_{i+1}\} \in E(G) \forall i \in \{1, \dots, k-1\}$
- The **length** of a walk is k (number of vertices in the walk)
- The walk is **closed** if $v_1 = v_k$
- A **trail** is a walk that does not use an edge twice
- A **path** is a walk for which $v_1, \dots, v_k$ are pairwise distinct



- $(1, 2, 3, 4, 2, 3)$ is a walk, but not a trail.
- $(1, 2, 3, 4, 2)$ is a trail, but not a path.
- $(1, 2, 3, 4)$ is a path.

Definition 1.10. Let G be a graph.

- A graph G is **connected** if there is a walk between any pair of its vertices
- A **(connected) component** of G is an inclusion wise maximal connected subgraph

1.1 Hamiltonian Cycles

Definition 1.11. A **Hamiltonian cycle** in a graph G is a spanning subgraph that is a cycle (isomorphic to C_n for some $n \in \mathbb{N}$)

If G has a Hamiltonian cycle, G is called **Hamiltonian**

Theorem 1.12 (Dirac's Theorem). *Let G be a graph of order $n \geq 3$ where the **order** of G is given by $|V(G)|$ and the **size** by $|E(G)|$. If every Vertex of G has degree at least $\frac{n}{2}$ then G is Hamiltonian*

Theorem 1.13 (Ore's theorem). *Let G be a graph of order $n \geq 3$. If for every pair $u, v \in V, u \neq v, u$ and v non-adjacent, we have $\deg(u) + \deg(v) \geq n$ then G is Hamiltonian*

Remark 1.14.

Ore $\implies$ Dirac

Proof.

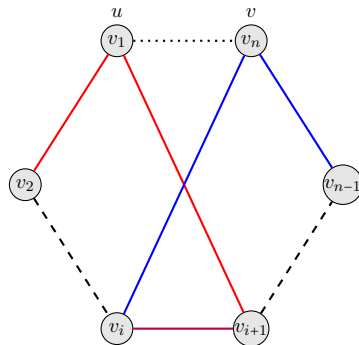
The Graph G is connected: suppose G is not connected. Take an arbitrary component of G and call it H . The order of H is $n - k$ if k vertices are not in $V(H)$. Now any $v \in V(H)$ has $\deg(v) \leq n - k - 1$ with equality if v is adjacent to every vertex in $V(H)$. Now take $u \notin V(H)$ so $\deg(u) \leq k - 1$. It follows that $u, v \in V(G), u \neq v$ and u, v are not adjacent so $\deg(u) + \deg(v) \geq n$ but $\deg(u) + \deg(v) \leq k - 1 + n - k - 1 = n - 2 < n$ $\nmid$

- G has a Hamiltonian cycle: By induction on the number of non-edges
- No non-edge $\implies G$ complete, hence Hamiltonian
- Assume G non complete $u, v \in V, u \neq v$, be non-adjacent. By induction the graph G' obtained from G by adding $e = \{u, v\}$ as an edge is Hamiltonian. Let C be a Hamiltonian cycle of G' . If C does not use $e \implies C$ is Hamiltonian cycle in G . Assume C uses e . Removing e from C yields a path from u to v

For $i \in \{2, \dots, n - 1\}$

- If v_i is a neighbour of u , color $\{v_{i-1}, v_i\}$ red
- If v_i is a neighbour of v , color $\{v_i, v_{i+1}\}$ blue

- Since $\deg(u) + \deg(v) \geq n$ and the path $v_1, \dots, v_n$ has $n - 1$ edges, there exists an edge $\{v_i, v_{i+1}\}$ which is both red and blue. Then $v_1, \dots, v_i, v_n, v_{n-1}, \dots, v_{i+1}$ is a Hamiltonian cycle in G .



□

1.2 Eulerian Trail

Definition 1.15. An **Eulerian trail** is a trail that passes through each edge of the graph exactly once. A graph with a closed Eulerian trail is called **Eulerian**

Theorem 1.16 (Euler's theorem). *Let G be a connected graph with at least 2 vertices. Then G has a closed Eulerian trail iff every vertex in G has even degree*

Proof.

" $\implies$ "

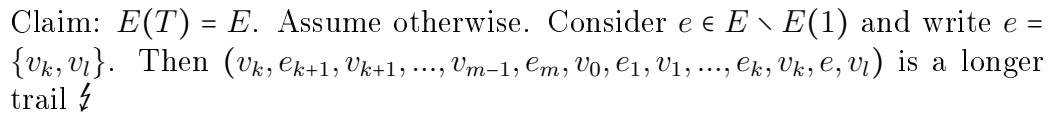
Let T be a closed Eulerian trail and fix an orientation on T . Then for every $v \in V(G)$, the incident edges can be classified as "ingoing" or "outgoing". This defines a bijection between in and outgoing edges of T , so $\deg(v)$ is even.

" $\impliedby$ "

Let $T = (v_0, e_1, v_1, \dots, e_m, v_m)$ be a longest trail. We will prove (i) $v_0 = v_m$ and (ii) $\{e_i : i \in \{1, \dots, m\}\} = E(G)$. (i)+(ii) $\iff T$ is closed Eulerian trail.

(i) If $v_0 \neq v_m$, then v_0 is incident to an odd number of edges in T . As $\deg(v_0)$ is even, there is $e \in E(G)$ incident to v_0 and not on T . We could thus append e on T $\nmid$ because T is longest trail.

(ii) Claim: $V(T) = V$. Assume otherwise there exists a vertex $v' \notin V(T)$ and $e = \{v_k, v'\} \in E(G)$ for some $k \in \{0, \dots, m\}$ (by connectedness). Then $(v', e, v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, \dots, v_{k-1}, e_k)$ is a longer trail hence $V(T) = V$



Theorem 1.17. *Let G be a connected graph with at least two vertices. Then G has an Eulerian trail iff there are precisely 0 or two vertices of odd degree*

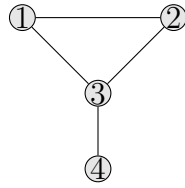
By the Euler's Theorem, we know that if there are 0 vertices of odd degree then there exists an Eulerian trail (in fact a closed Eulerian trail), and if there exists a closed Eulerian trail, then all vertices have even degree.

1.3 Datastructures

Definition 1.18. The **adjacency matrix** of G is the matrix $A = (a_{ij})$ with

The **adjacency list** of G stores a label for each vertex, and a list of (pointers to) its neighbours.

Example 1.19.



$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

- v_1 : v_2, v_4
- v_2 : v_1, v_3, v_4
- v_3 : v_2
- v_4 : v_1, v_2

Remark 1.20.

Memory requirement:

Adjacency matrix: $\theta(n^2)$

Adjacency list: $\theta(n + m)$, where m is the number of edges

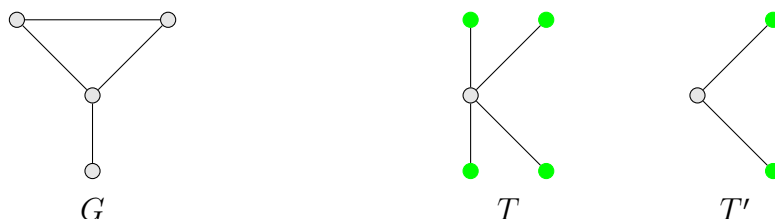
2 Trees

Definition 2.1. A **tree** is a graph that is connected and does not contain a cycle (a graph "contains a cycle" if it has a subgraph isomorphic to C_n for some $n \geq 3$).

A **forest** is a graph that does not contain a cycle.

A **leaf** is a vertex of a tree with degree 1.

Example 2.2.



- G is not a tree nor a forest
- T, T' are trees and forests
- $T \cup T'$ is a forest but not a tree
- All green vertices are leafs, all white vertices are not leafs

Lemma 2.3. Every tree T with $n \geq 2$ vertices has at least one leaf.

Proof. We iteratively construct a path $v_1, \dots, v_n$ as follows:

Start with any edge $\{v_1, v_2\} \in E(T)$. Assume that we have constructed $v_1, \dots, v_k$. To construct v_{k+1} , we may assume that v_k is not a leaf in T (or we are done). Otherwise v_k has a neighbour $v_{k+1} \neq v_{k-1}$. $v_{k+1} \neq v_i$ because T is a tree and otherwise we would get a cycle $(v_i, \dots, v_k, v_{k+1} = v_k)$. Add v_{k+1} to the path and continue finite many steps (because there are only finitely many vertices). The algorithm just stated ends at a leaf. $\square$

Theorem 2.4. Let G be a graph with n vertices. The following are equal:

- G is a tree
- G is connected with $n - 1$ edges
- G is connected and removing an edge disconnects it
- G has exactly one path between each pair of distinct vertices

(v) G has no cycle, but adding an edge creates a cycle

Proof. "1. $\implies$ 2."

Induction on n :

$n = 1$ is trivial

Let $n \geq 2$. By Lemma 2.3, G has a leaf v . Removing v and the unique edge incident to v yields a tree G' . By induction G' has $n - 2$ edges so G has $n - 1$ edges.

"2. $\implies$ 3."

Suppose there is an edge $e \in E(G)$ and the resulting graph by removing e is still connected. Keep removing as long as the resulting graph is still connected. Let G' be the resulting graph. Observe that G' is a spanning subgraph of G . G' does not contain a cycle as otherwise we could remove one edge from the cycle. By "1. $\implies$ 2.", applied to G' , we know that G' has $n - 1$ edges $\nmid$ because G had $n - 1$ edges and we removed at least one.

"3. $\implies$ 4."

Suppose that there is a pair with $u, v \in V(G)$ s.t. there are two distinct paths u_i, v_i between them. Let i_0 be the maximum index s.t. $u_i = v_i : \forall i \in \{1, \dots, i_0\}$. The edge $\{v_{i_0}, v_{i_0+1}\}$ does not lie on the path u_i . The graph obtained by removing $\{v_{i_0}, v_{i_0+1}\}$ is connected as for any walk using this edge we can use u_i to go to u to reach v_{i_0} or v to reach v_{i_0+1} .

"4. $\implies$ 5."

If G had a cycle, then there would exist two paths between each pair of neighbouring vertices on the cycle (clockwise + counter clockwise).

Adding an edge creates a cycle: Let $u, v \in V(G)$ non adjacent. There is one path between the two. Now adding an edge between u, v creates a second path $\nmid$

"5. $\implies$ 1."

We only need to show that G is connected. Suppose u, v are not connected by a walk. Creating an edge between u, v does not result in a cycle because for being isomorphic to C_n one needs at least two walks between each of the vertices in the cycle but by assumption the only walk connecting u, v is the newly added edge $\nmid$ $\square$

Corollary 2.5. *Every connected graph G has a spanning tree.*

Proof.

Consider graphs on n vertices, now do induction on the number of edges m . As G is connected, $m \geq n - 1$. If $m = n - 1 \implies G$ is a tree. If $m > n - 1 \implies G$ is not a

tree. By Theorem 2.4 we can remove edges s.t. the resulting graph is connected. By induction it has a spanning tree, which is then also a spanning tree of G . $\square$

Theorem 2.6 (Matrix-tree theorem). *Let G be a graph with vertices $\{1, \dots, n\}$, edges $\{e_1, \dots, e_m\}$.*

Let $Q = (q_{ij})$ with $q_{ij} = \begin{cases} -1 & \{i, j\} \in E \\ 0 & i \neq j : \{i, j\} \notin E \\ \deg(i) & i = j \end{cases}$ Let Q_{ij} be the matrix obtained

from Q by removing the i -th row and the j -th column. Then the number of labelled spanning trees of G is $|\det(Q_{ij})|$, for any choice of i, j .

Corollary 2.7 (Cayley's formula). *The number of labelled spanning trees of K_n is n^{n-2} if $n \geq 2$.*

Proof. Define Q as in Theorem 2.6.

$$Q_{11} = \begin{pmatrix} n-1 & 1 & \dots & 1 \\ 1 & \ddots & & \vdots \\ \vdots & & \ddots & 1 \\ 1 & \dots & 1 & n-1 \end{pmatrix} \rightarrow \bar{Q}_{11} = \begin{pmatrix} n-1 & -1 & -1 & \dots & -1 \\ -n & n & 0 & \dots & 0 \\ -n & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & 0 \\ -n & 0 & \dots & 0 & n \end{pmatrix} \rightarrow \hat{Q}_{11} = \begin{pmatrix} 1 & \star & \dots & \star \\ 0 & n & & \vdots \\ \vdots & & \ddots & \star \\ 0 & \dots & 0 & n \end{pmatrix}$$

$$\implies |\det(Q_{11})| = |\det(\hat{Q}_{11})| = n^{n-1-1} = n^{n-2} \quad \square$$

3 Planar Graphs

Definition 3.1. A **planar drawing** of a graph G consists of injective continuous functions

$$\phi_v : V(G) \rightarrow \mathbb{R}^2 \qquad \phi_e : [0, 1] \rightarrow \mathbb{R}^2$$

s.t.:

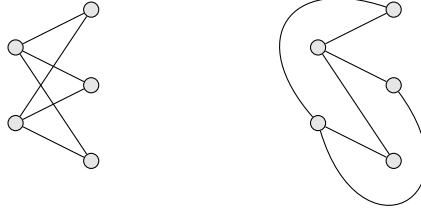
- $\{\phi_{\{u,v\}}(0), \phi_{\{u,v\}}(1)\} = \{\phi_u(u), \phi_v(v)\}$ for all $\{u, v\} \in E(G)$
- The sets $\phi_e((0, 1))$ for each $e \in E(G)$ are pairwise disjoint and disjoint from $\text{Im}(\phi_v)$ (i.e. no crossings)

A graph G is called **planar** if it has a planar drawing.

A **plane graph** is a planar graph, together with a fixed planar drawing.

Example 3.2.

- Trees
- K_4
- $K_{2,3}$



Definition 3.3. A **face** of a plane graph G is a maximal connected part of $\mathbb{R}^2 \setminus (\text{Im}(\phi_v(u)) \cup \text{Im}(\phi_e))$. Every plane graph has exactly one unbounded face called the **outer face**. All other faces are called **inner faces**. The **boundary** of a face are the vertices and edges of G incident with it. A plane graph is called **planar triangulation** if every face is bounded by a cycle of length precisely 3 (this includes the outer face).

Theorem 3.4 (Euler's formula). *Let $G = (V, E)$ be a connected planar graph. Then the number of faces $|F|$ of G is $|F| = |E| - |V| + 2$.*

Proof. Induction on $|E|$.

Since G is connected, we have $|E| \geq |V| - 1$ so we start the induction at $|E| = |V| - 1 \implies G$ is a tree. Since a tree does not contain a cycle, we have $|F| = |E| - |V| + 2$. Now let $|E| > |V| - 1$. Then G is not a tree, so G contains a cycle C . Choose an edge e on C and remove it to obtain a plane graph G' . e separates two faces in G , which fuse to a single face in G' . By induction, the number of faces $|F'|$ in G' is $|F| - 1 = |F'| = |E| - 1 - |V| + 2 \implies |F| = |E| - |V| + 2$. $\square$

Lemma 3.5. *For every plane graph $G = (V, E)$ with $|V| \geq 3$, we have $|E| \leq 3|V| - 6$.*

Proof. Wlog G is connected. Consider a planar drawing of G and let F be the set of faces. Every face is bounded by at least 3 edges and each edge bounds at most 2 faces. By Euler's formula 3.4, we obtain $\frac{2}{3}|E| \geq |F| = |E| - |V| + 2 \implies |E| \leq 3|V| - 6$. $\square$

Remark 3.6. This means that the number of edges in a planar graph grows only linear proportional to the number of vertices.

Example 3.7.

- K_5 is not planar (10 edges) due to Theorem 3.4
- $K_{3,3}$ is not planar (without proof)

Definition 3.8. A **subdivision** of a graph G is a graph obtained from G by repeatedly subdividing some of its edges.

Theorem 3.9 (Kuratowski's theorem). *A graph G is planar iff G contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.*

Remark 3.10. A subdivision of a graph is planar iff the graph is planar

Definition 3.11. Let G be a graph. The graph obtained from G by contracting $e = \{u, v\} \in E(G)$ denoted G/e , is the graph obtained from G by deleting u, v , inserting a new vertex w and making w adjacent to all vertices in $N_G(u) \cup N_G(v)$. A graph G' is a **minor** of G if G' can be obtained from G by a sequence of vertex deletions, edge deletions and edge contractions.

Remark 3.12. If G' is a minor of G then G' planar $\iff G$ is planar.

Theorem 3.13 (Wagner's theorem). *A graph G is planar iff $K_{3,3}$ and K_5 are not a minor of G .*

4 Flows, Cuts, Connectivity

Definition 4.1. A **directed graph** is a pair (V, E) with $E \subseteq (V \times V) \setminus \{(x, x) : x \in V\}$. Write $\delta_G^+(v)$ for edges starting in v and $\delta_G^-(v)$ for edges ending in v .

Definition 4.2. A **flow network** is a tuple (G, s, t, c) where G is a **directed** graph, $s, t \in V(G)$, $s \neq t$ and $c : E \rightarrow \mathbb{R}_0^+$.

- s is called a **source**
- t is called a **target**
- $\forall e \in E(G) : c(e)$ is the **capacity** of e

A **flow** in a flow network is a function $f : E \rightarrow \mathbb{R}_0^+$ s.t.

- $f(e) \leq c(e)$ for all $e \in E(G)$
- $\forall v \notin \{s, t\}, \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$ (i.e. what flows in flows out)
- $\forall e \in \delta^-(s) \cup \delta^+(t) : f(e) = 0$ (i.e. nothing flows into a source or out of a target)

The **value** of a flow f is $\text{val}(f) = \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$

Remark 4.3. For $v \in V$, write $f(\delta^+(v)) = \sum_{e \in \delta^+(v)} f(e)$; $f(\delta^-(v)) = \sum_{e \in \delta^-(v)} f(e)$

Definition 4.4. A **cut** in a graph/directed graph G is a partition of $V(G)$ into two non-empty parts A and B (i.e. $V(G) = A \dot{\cup} B$ and $A, B \neq \emptyset$). For $s, t \in V(G)$, $s \neq t \implies (A, B)$ is a (s, t) -cut if $s \in A, t \in B$.

If (S, T) is a (s, t) -cut, write $\delta^+(S) = \{e \in E(G) : e \text{ start in } S \text{ ends in } T\} = \delta^-(T)$ and $\delta^-(S) = \{e \in E(G) : e \text{ starts in } T \text{ and ends in } S\} = \delta^+(T)$.

The **capacity** of the cut is $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = c(\delta^+(S)) = c(\delta^-(T))$.

Theorem 4.5 (max flow-min cut theorem). *Let (G, s, t, c) be a flow network. Then $\max\{\text{val}(f) : f \text{ flow}\} = \min\{c(S, T) : (S, T) \text{ is a } (s, t)\text{-cut}\}$.*

Lemma 4.6. *Let f be a flow, (S, T) be a (s, t) -cut. Then $\text{val}(f) = f(\delta^+(S)) - f(\delta^-(S))$.*

Proof.

Claim: $\forall A \subset V : f(\delta^+(A)) - f(\delta^-(A)) = \sum_{v \in A} f(\delta^+(v)) - f(\delta^-(v)) = (*)$

The sum $f(\delta^+(A)) - f(\delta^-(A))$ does not account for edges $e \in E(G)$ with start and end in A . On the right side, $f(e)$ is counted twice, once with positive and once with negative sign, so the sum vanishes.

Then $\text{val}(f) = f(\delta^-(t)) \stackrel{(1)}{=} \sum_{v \in T} f(\delta^-(v)) - f(\delta^+(v)) \stackrel{(*)}{=} f(\delta^-(T)) - f(\delta^+(T)) = f(\delta^+(S)) - f(\delta^-(S))$

(1) holds because $f(\delta^+(t)) = 0$ (t is a sink) and $\forall v \notin \{s, t\} : f(\delta^-(v)) - f(\delta^+(v)) = 0$ (by definition). $\square$

Corollary 4.7. *Let f be a flow, (S, T) an (s, t) -cut.*

$\text{val}(f) \leq c(\delta^+(S))$

Thus $\max\{\text{val}(f) : f \text{ flow}\} \leq \min\{c(S, T) : (S, T) \text{ (s, t)-cut}\}$.

Proof. Theorem 4.5

max flow $\leq$ min cut due to Corollary 4.7

Let f maximum flow (i.e. a flow for which $\text{val}(f)$ is maximal). Let $c' : E(G) \rightarrow \mathbb{R}_0^+$ where $c'(e) = c(e) - f(e)$. Let $S \subset V(G)$ be defined as the set of reachable vertices¹ according to the following rules:

- $s \in S$
- $x \in V(G)$ is in S (i.e. reachable) if there is a walk (in the underlying directed graph) from s to x traversing $e_1, \dots, e_k$ s.t.
 - (i) $c'(e_i) > 0$ ($\iff f(e_i) < c(e_i)$) if e_i is traversed towards x .
 - (ii) $c'(e_i) < c(e_i)$ ($\iff f(e_i) > 0$) if e_i is traversed towards s .

If $S = V(G)$, we can improve f as follows:

Let $e_1, \dots, e_k$ be a walk from s that makes t reachable, and e_i is traversed in orientation $\sigma_i \in \{\pm 1\}$, then $\tilde{f}(e_i) = f(e_i) + \sigma_i \varepsilon$, for $i \in \{1, \dots, k\}$ and $\tilde{f}(e_i) = f(e_i)$ otherwise. This defines a flow $\tilde{f}$ with $\text{val}(\tilde{f}) > \text{val}(f)$ for $\varepsilon > 0$ sufficiently small $\nexists$. Thus $t \notin S$. Set $T = V \setminus S \neq \emptyset$.

- $\forall e \in \delta^+(S) \quad c'(e) = 0$, i.e. $f(e) = c(e)$
 $e = \{x, y\}$ with $x \in S, y \in T$. If $c'(e) \neq 0 \implies e$ satisfies (i) $\implies y$ reachable $\nexists$

¹These are the vertices where the flow "is not maximal yet" but has a walk to s .

- $\forall e \in \delta^-(S) \quad c'(e) = c(e) \iff f(e) = 0$
- $e = \{y, x\}$ with $x \in S, y \in T$. If $c'(e) \neq c(e) \implies e$ satisfies (ii) $\implies y$ reachable $\nmid$
- $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = \sum_{e \in \delta^+(S)} f(e) - \sum_{e \in \delta^-(S)} f(e) = f(\delta^+(S)) - f(\delta^-(S)) \stackrel{(4.6)}{=} \text{val}(f) \implies \text{val}(f) \geq c(S, T) \geq \min(s, t)\text{-cut}$ $\square$

4.1 Connectivity

Definition 4.8. A graph G is **k -vertex connected** ($k \in \mathbb{N}$) if it has at least $k + 1$ vertices and whenever we remove up to $k - 1$ vertices, the resulting graph is connected.

A graph G is **k -edge connected** if whenever we remove up to $k - 1$ edges, the resulting graph is connected.

The **vertex connectivity (edge connectivity)** is the largest $k \in \mathbb{N}$ s.t. G is k -vertex/edge connected.

Remark 4.9. G graph, k vertex connectivity, λ edge connectivity, δ minimal degree of G , then $k \stackrel{(i)}{\leq} \lambda \stackrel{(ii)}{\leq} \delta$

Proof. (i) If we have a set of λ edges whose removal makes the res. graph disconnected, then deleting one endpoint for each edge also disconnects the graph.

(ii) Delete all edges incident to a vertex of minimal degree, this disconnects the graph. $\square$

Lemma 4.10. G directed graph, $s, t \in V, s \neq t$. Then the maximal number of edge-disjoint (s, t) -paths (path which do not have any common edges) is equal to the size of a minimal (S, T) -cut.

Proof. "Max flow-min cut theorem" 4.5 with $\forall e \in E \quad c(e) = 1$. $\square$

Theorem 4.11 (Menger's theorem for edge-connectivity). Let G be a graph, $k \in \mathbb{N}$. G is k -edge-connected iff there are k edge-disjoint paths between any pair of vertices in G .

Proof.

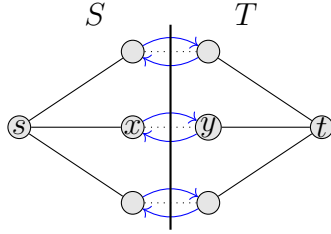
" $\iff$ " Removing $k - 1$ edges will keep one edge intact, so the resulting graph is connected, hence G is k -edge connected.

" $\implies$ " Let $s, t \in V, s \neq t$. Let G' be the graph obtained from G by replacing each edge by a double-edge (see graph(1)).

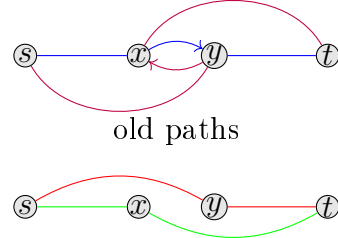
Claim: Any (s, t) -cut in G' has at least size k .

If (S, T) is an (s, t) -cut of size $\leq k - 1$, then G has at most $k - 1$ edges between S and T . Removing them disconnects the graph $\nmid G$ is k -edge connected. By Lemma 4.10, there are k -edge disjoint paths $p_1, \dots, p_k$ from s to t in G' .

Suppose $x, y \in V$ s.t. p_i uses $\{x, y\}$ and p_j uses $\{y, x\}$. Replace p_i and p_j by the path following p_i to x and continue via p_j and the path following p_j to y and continue via p_i . Repeat for each such edge. In each step, the sum of the lengths of the paths decreases so the process has to terminate. Then the underlying k -undirected paths in G are edge disjoint (see graph(2)).



graph(1)



new paths

graph(2)

□

Theorem 4.12 (Menger's theorem for vertex-connectivity). *Let G be a graph with at least $k + 1$ vertices ($k \in \mathbb{N}$). Then G is k -vertex connected iff there are at least k internally disjoint paths (start and end are the only common vertices) between any pair of distinct vertices of G .*

Proof. " $\Leftarrow$ " As before (Proof of Theorem 4.11).

" $\implies$ " Let $a, b \in V$ be distinct and $v_1, \dots, v_n$ be the remaining vertices of G . Consider the following directed Graph G' (see graph(1)):

- $V(G') = \{s, t, v'_1, \dots, v'_n, v''_1, \dots, v''_n\}$
- $E(G') = \{\{v'_i, v''_i\} : i \in \{1, \dots, n\}\} \cup \{\{v''_i, v'_j\} : \{v_i, v_j\} \in E(G)\} \cup \{\{v''_j, v'_i\} : \{v_i, v_j\} \in E(G)\} \cup \{\{s, v_i\} : \{a, v_i\} \in E(G)\} \cup \{\{v''_i, t\} : \{b, v_i\} \in E(G)\} \cup \{\{s, t\} \text{ if } \{a, b\} \in E(G)\}$ where we call elements of $\{\{v'_i, v''_i\} : i \in \{1, \dots, n\}\}$ red edges.

Let k_0 be the minimum size of an (s, t) -cut in G' . If $k_0 \geq k$, G' has k -edge disjoint paths from s to t . Since no red edge is contained in two of those, this corresponds to k internally disjoint paths from a to b in G : each path is of the form $(s, v'_1, v''_1, \dots, v'_m, v''_m, t)$ and corresponds to $(a, v_i, \dots, v_m, b)$ in G .

Now assume $k_0 < k$ and consider an (s, t) -cut (S, T) of G' of size k_0 .

Claim: We may assume that every edge in $\delta^+(S) \setminus \{(s, t)\}$ is red:

If an edge entering v'_i is in $\delta^+(S)$, move v'_i from T to S without changing the size of the cut. Similarly if an edge leaving v''_i is in $\delta^+(S)$, move v''_i to T . If $\{s, t\} \notin E(G)$, then removing the k_0 vertices of G that correspond to red edges in $\delta^+(S)$ disconnects G (no path from s to t in G' after removal $\implies$ none from a to b in G) $\nmid$.

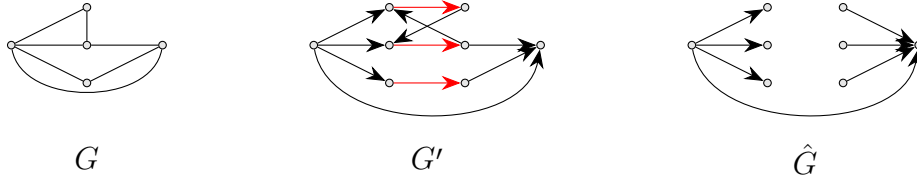
If $\{s, t\} \in E(G)$, proceed as follows:

Let A be the set of vertices in G corresponding to red edges in $\delta^+(S)$ in G' ($|A| = k_0 - 1$). Let $\hat{G}$ be the graph obtained by removing A .

If $\hat{G}$ not connected $\nmid G$ is k -vertex connected.

Thus $\hat{G}$ connected. Since G has $\geq k + 1$ vertices either a or b has an additional neighbour.

Wlog a does $\implies \hat{G} - a$ is not connected $\nmid G$ is k -vertex connected (see graph(2)).



□

Definition 4.13. A graph is **bipartite** with parts A, B if $V(G) = A \dot{\cup} B$ and every edge in G has one end in A and one in B .

Example 4.14. • $K_{a,b}$

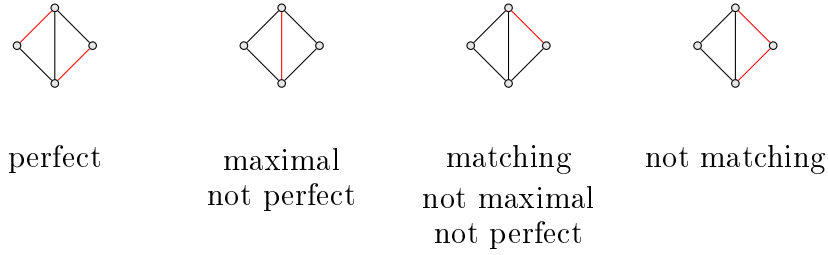
• $Q_d, d \geq 1$

Definition 4.15. A **matching** in a graph G is a set $M \subset E(G)$ s.t. no two edges in M are incident.

A matching M is **perfect** if every vertex in G is incident to an edge in M .

A maximal matching is a matching where no edge from $E(G)$ can be added to the matching without violating the matching property.

Example 4.16.



Theorem 4.17 (Hall's marriage theorem). *Let $G = (V, E)$ be a bipartite graph with parts A, B . Then G has a perfect matching iff $\forall P \subseteq A, |N_G(P)| \geq |P|$ and $\forall P \subseteq B, |N_G(P)| \geq |P|$.*

Proof. " $\implies$ " Assume (Wlog) $\exists P \subseteq A$ s.t. $|N_G(P)| < |P|$. Let M be a perfect matching. Then $\forall v \in P$ the other endpoint of the matching edge incident to v is in $N_G(v) \subset N_G(P)$, and these vertices are distinct $\implies |N_G(P)| \geq |P|$ $\nmid$.

" $\impliedby$ " Note that the condition implies $|A| = |B|$, since $|B| \geq |N_G(A)| \geq |A|$ and vice versa. Define an auxiliary graph $G' = (V', E')$ with:

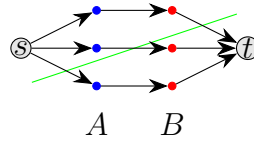
- $V' = V \cup \{s, t\}$
- $E' = \{\{s, a\} : a \in A\} \cup \{\{b, t\} : b \in B\} \cup \{\{a, b\} : a \in A, b \in B\}$
- $\forall e \in E : c(e) = 1$

Every matching M in G defines an integral (s, t) -flow ($f(e) \in \{0, 1\}$) in G' with $\text{val}(f) = |M|$ and vice versa.

Let (S, T) be an (s, t) -cut. Let $A_S = A \cap S, A_T = A \cap T, B_S = B \cap S, B_T = B \cap T$. It follows $|\delta^+(S)| \geq |A_T| + |B_S| + |N_{A_S}(B_T)| \geq |N_{A_T}(B_T)| + |B_S| + |N_{A_S}(B_T)| = |N_A(B_T)| + |B_S| \stackrel{\text{condition}}{\geq} |B_T| + |B_S| = |B| = |A|$.

Thus by maxflow-mincut, $\max_{f \text{ flow}} \text{val}(f) = \max_{f \text{ integral flow}} \text{val}(f) \geq |B|$.

Therefore there exists a matching in G of size $\geq |B|$, i.e. a perfect matching.



□

5 Coloring

Definition 5.1. Let $k \in \mathbb{N}$. A (proper) **k -vertex-coloring** (k -coloring) of a graph G is a function $c : V(G) \rightarrow \{1, 2, \dots, k\}$ such that for every edge $\{u, v\} \in E(G)$, we have $c(u) \neq c(v)$. The **chromatic number** $\chi(G)$ of a graph G is the smallest integer k such that G has a proper k -coloring. We call G **k -colorable** if it has a k -vertex-coloring.

Write $\Delta(G)$ for the maximum degree of a vertex in G .

Lemma 5.2. *Let G be a graph. Then $\chi(G) \leq \Delta(G) + 1$.*

Proof. Order the vertices of G as $v_1, v_2, \dots, v_n$. We will color the vertices in this order. When we come to color v_i , it has at most $\Delta(G)$ neighbors, so there are at most $\Delta(G)$ colors that are forbidden for v_i . Since we have $\Delta(G) + 1$ colors available, there is always at least one color that we can use to color v_i . $\square$

Remark 5.3. The bound $\chi(G) \leq \Delta(G) + 1$ is tight with equality for exactly the complete graphs K_n and the odd cycles C_{2k+1} .

Lemma 5.4. *Let G be a connected graph. If there is $v \in V(G)$ such that $\deg_G(v) < \Delta(G)$, then G has a coloring with $\Delta(G)$ colors.*

Proof. Let $w \in V(G)$ such that $\deg_G(w) < \Delta(G)$. Fix a spanning tree T of G rooted at w . Let $v_1, \dots, v_n$ be the vertices of G listed in the way such that each non-root vertex precedes its parent (and the root is the last vertex, i.e., $v_n = w$). Color the vertices one by one, starting at v_1 . Suppose that $v_1, \dots, v_{i-1}$ are already colored. The vertex v_i is adjacent to at most $\Delta(G) - 1$ vertices among $v_1, \dots, v_{i-1}$: if $i < n$, then the parent of v_i is not among these vertices, and if $i = n$, then $v_i = w$ has degree at most $\Delta(G) - 1$ by assumption. In both cases, there thus exists a color not assigned to any neighbor of v_i , and we can assign it to v_i . This yields vertex coloring of G using at most $\Delta(G)$ colors. $\square$

Theorem 5.5 (Brooks' Theorem). *Let G be a connected graph. If G is neither an odd cycle nor a complete graph, then $\chi(G) \leq \Delta(G)$.*

Proof. $\Delta(G) = 0$: Then G is a single vertex and therefore complete.

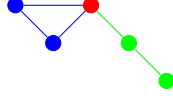
$\Delta(G) = 1$: Then G is a single edge and therefore complete.

$\Delta(G) = 2$: Then G is a cycle. By assumption, G is an even cycle, so $\chi(G) = 2 = \Delta(G)$.

Let $\Delta(G) \geq 3$

By Lemma 5.4, we may assume that G is $\Delta(G)$ -regular ($\forall v \in V(G)$, $\deg_G(v) = \Delta(G)$).

- (i) $\exists v \in V(G)$ such that $G-v$ is disconnected. Then $G-v$ has at least two components. Let $C_1, \dots, C_k$ be the components of $G-v$, and let $G_i = G[V(C_i) \cup \{v\}]$. G_i is connected, $\Delta(G_i) = \Delta(G)$ and since $\deg_{G_i}(v) < \deg_G(v) = \Delta(G) \implies$ (by Lemma 5.4) we can color G_i with $\Delta(G)$ colors. Fix such a coloring and then permute the colors s.t. v always has the same color. Then this is a coloring of G with $\Delta(G)$ colors.

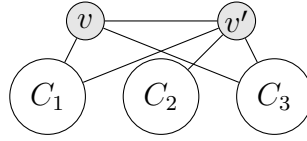


- (ii) Wlog G is connected.

Assume not (i) and $\exists v, v' \in V(G)$ s.t. $G-v-v'$ is disconnected.

Let $C'_1, \dots, C'_k$ be the components of $G-v-v'$. Let $G_i = G[V(C'_i) \cup \{v, v'\}]$. Let G'_i be the graph obtained from G_i by adding $\{v, v'\}$ if it is not already there.

Observe that $\forall i \in \{1, \dots, k\}$, v has a neighbour in C_i (say v'_i).



$$\deg_{G'_i}(v) = \begin{cases} \deg_G(v) - |\bigcup_{j \neq i} N_{C_j}(v)| & \{v, v'\} \in E(G) \\ \deg_G(v) + 1 - |\bigcup_{j \neq i} N_{C_j}(v)| & \{v, v'\} \notin E(G) \end{cases} \leq \begin{cases} \Delta(G) - (k-1) & \{v, v'\} \in E(G) \\ \Delta(G) - (k-2) & \{v, v'\} \notin E(G) \end{cases}$$

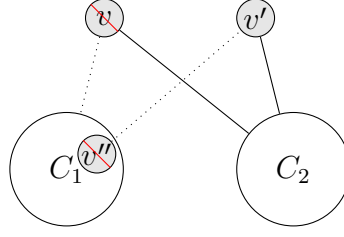
Same goes for $\deg_{G'_i}(v')$.

If $\deg_{G'_i}(v) < \Delta(G)$ or $\deg_{G'_i}(v') < \Delta(G)$, then we can $\Delta(G)$ -color G_i by Lemma 5.4.

If $G'_1, \dots, G'_k$ have a $\Delta(G)$ -coloring $\implies$ can combine them to a $\Delta(G)$ -coloring of G .

Suppose $\exists i \in \{1, \dots, k\}$ s.t. $\deg_{G'_i}(v) = \deg_{G'_i}(v') = \Delta(G)$. Wlog $i = 2$. Then $k = 2, \{v, v'\} \notin E(G)$ ($\deg_{G'_i}(v) < \Delta(G)$ if $k \geq 3$) and $|N_{C_1}(v)| = |N_{C_2}(v')| = 1$. Let $v'' \in V(C_1)$ be the neighbour of v'

Observation: $\bar{G} = G-v-v''$ is not connected. $\bar{G}$ has components $D_1 := C_1-v''$ and $D_2 = [V(C_2) \cup \{v'\}]$. v has $\Delta(G)-1$ neighbours in D_2 and v'' has $\Delta(G)-1$ neighbours in D_1 . Use the above coloring argument on $G-v-v''$.



(iii) Wlog G is connected.

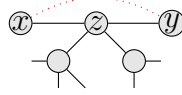
Assume not (i) and not (ii). Then $\forall v, v' \in V(G)$, $G - v - v'$ is connected.

G not complete $\implies \exists x, x' \in V(G), x \neq x'$, non adjacent. Let P be a shortest path from x to x' . Let z be the neighbour of x on P and y be the neighbour of z on P distinct from x . As P is a shortest path, x and y are nonadjacent. Fix a spanning tree of $G - x - y$ rooted at z . Set $v_1 = x, v_2 = y$ and $v_3, \dots, v_n$ is a listing of the remaining vertices of G s.t. every vertex in the tree precedes its parents. In particular $v_n = z$. Now we give the same color to v_1 and v_2 .

Suppose $v_1, v_2, \dots, v_{i-1}$ are colored already.

If $i < n \implies v_i$ is adjacent to at least one vertex in $\{v_{i+1}, \dots, v_n\}$ namely its parent. So v_i has at most $\Delta(G) - 1$ neighbours that already have a color, so we have 1 color left.

For $i = n, z$ has two neighbours with same color (x, y) , so again there is a color not present among the neighbours of z , which we assign to z .



□

Theorem 5.6 (4-Color Theorem). *Every planar graph is 4-colorable.*

Theorem 5.7 (5-Color Theorem). *Every planar graph is 5-colorable.*

Proof. Induction on $n = |V(G)|$. Fix a plane drawing of G .

$n = 1, 2$: trivial.

Let $n \geq 3$:

Suppose $\exists v \in V(G)$ $\deg_G(v) \leq 4$. Then color $G - v$ by ind. with 5 colors. The neighbours of v use at most 4 colors, so there is a color we can assign to v .

Now assume that G is maximal.

From the proof of Lemma 3.4, it follows that G is a triangulation. By Ex 3.2 $\exists w \in V(G)$ s.t. $\deg_G(w) = 5$. Let $v_1, \dots, v_5$ be the neighbours of w , listed in cyclic ordering around w . Then $\{v_1, v_3\}$ or $\{v_2, v_4\}$ is not an edge in $E(G)$, Wlog $\{v_1, v_3\}$.

Let G' be the plane graph obtained by removing w, v_1, v_3 and inserting w' at the position of w . Join every neighbour u of v_1 in G to w' by inserting an edge along $\{u, v_1\}$ and then along $\{v_1, w\}$ (same for the neighbours of v_3 but put only one neighbour of both v_1, v_3). $|V(G')| = n - 2$. G' is planar. By induction, there exists a 5-coloring $c' : V(G') \rightarrow \{1, \dots, 5\}$ of G' .

$$\text{Let } c : V(G) \rightarrow \{1, \dots, 5\} \text{ with } c(v) = \begin{cases} c'(v), & v \in V(G) \setminus \{w, v_1, v_3\} \\ c(v_1) = c'(w') \\ c(v_3) = c'(w') \\ c(w) \text{ is any color } \notin \{c(v_i) : i = 1, \dots, 5\} \end{cases}$$

Then c is a 5-coloring of G . □

Definition 5.8. The **clique number** of a graph G , denoted $\omega(G)$, is the order of the largest complete subgraph of G .

Remark 5.9. Clearly, $\omega(G) \leq \chi(G) \leq \Delta(G) + 1$.

Theorem 5.10. Let $k \geq 2$. There exists a graph G with $\omega(G) = 2$ and $\chi(G) > k$.

Definition 5.11. The **girth** of a graph G is the length of the shortest cycle in G .

Theorem 5.12. For every $k \geq 1$, there exists a graph G with $g(G) > k$ and $\chi(G) > k$.

Proof. postponed □

Remark 5.13. $5.12 \implies 5.10$ as $g(G) > 3 \implies G$ has no triangles $\implies \omega(G) \leq 2$.

Conjecture 5.14 (Reed). $\chi(G) \leq \lceil \frac{\omega(G) + \Delta(G) + 1}{2} \rceil$

5.1 PerfectGraphs

Definition 5.15. A graph G is **perfect** if for every induced subgraph G' of G , we have $\chi(G') = \omega(G')$.

Example 5.16. complete graphs and bipartite graphs are perfect.

Theorem 5.17 (Weak perfect graph theorem). A graph G is perfect if and only if $\overline{G}$ is perfect.

($\overline{G}$ is the complement of G which is the graph with the same vertex set as G and two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G)

Theorem 5.18 (Strong perfect graph theorem). A graph G is perfect if and only if G does not contain an odd cycle of length at least 5 or the complement of an odd cycle of length at least 5 as an induced subgraph.

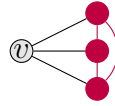
Definition 5.19. A graph is **chordal** if it does not contain an induced cycle of length at least 4.

Remark 5.20. Induced subgraphs of chordal graphs are chordal.

Theorem 5.21. *Every chordal graph is perfect.*

Proof. Use Theorem 5.17. □

Definition 5.22. A vertex v in a graph G is **simplicial** if $N_G(v)$ induces a complete graph.



Lemma 5.23. *Let G be a graph. If G is chordal, then it has a simplicial vertex.*

Proof. Induction on $n := |V(G)|$. Statement is true for $n = 1$ and if G is complete. Now $n \geq 2$ and G not complete.

Choose $U \subseteq V(G)$ of maximal size s.t. $G[U]$ (= subgraph induced by the vertices in U) is connected and $U \cup N_G(U) \neq V(G)$ ($N_G(U) = \bigcup_{u \in U} N_G(u) \setminus U$).

If $u, u' \in V(G)$ are non-adjacent, $u \neq u'$, then $U = \{u\}$ satisfies the above condition. Let $W = V(G) \setminus (U \cup N_G(U))$, then $W \neq \emptyset$. Every $x \in N_G(U)$ is adjacent to all vertices in W , otherwise we could add x to U ~~by~~ maximality of U . $G[W]$ is chordal and by induction W has a simplicial vertex v .

Claim: v is simplicial in G .

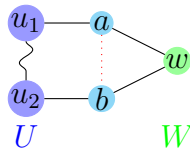
We show that $N_G(v) := N(U)$ is a clique.

Suppose there are $a, b \in N(U)$, a and b non-adjacent, $a \neq b$. Let $w \in W \implies w$ is adjacent to both a and b . Let P be a shortest path from a to b with inner vertices in U (exists since $G[U]$ is connected and both a, b have neighbours in U).

Then concatenating P with $(b, \{b, w\}, w, \{a, w\}, a)$ yields a cycle in G that does not have a chord, the cycle has length ≥ 4 ~~by~~ G chordal.

$\implies N(U)$ is a clique.

$\implies N_G(v) = N(U)$ is a clique $\implies v$ is simplicial in G .



□

Theorem 5.24. *Let G be a graph. Then G is chordal iff G has a perfect elimination ordering, i.e. an ordering $v_1, \dots, v_n$ of $V(G)$ s.t. $\forall i \in \{1, \dots, n\}$, the neighbours of v_i in $\{v_{i+1}, \dots, v_n\}$ induce a complete graph.*

Proof. " $\implies$ " Let v_1 be a simplicial vertex in G (exists by Lemma 5.23). Proceed inductively on $G - v_1$ (which is chordal).

" $\impliedby$ " Let C be an induced cycle of G of length ≥ 3 . Let v_i be a PEO. Let i be minimal s.t. $v_i \in V(C)$. Then the two neighbours of v_i on C are v_{i+j} and v_{i+l} for some $j, l > 0$. Since the neighbours of v_i in $\{v_{i+1}, \dots, v_n\}$ induce a complete graph, v_{i+j} and v_{i+l} are adjacent, so $|V(C)| = 3$. Thus G is chordal. $\square$

5.2 EdgeColoring

Definition 5.25. Let $k \in \mathbb{N}$. A (proper) **k -edge-coloring** of a graph G is a function $c : E(G) \rightarrow \{1, 2, \dots, k\}$ such that no two adjacent edges have the same color.

The **chromatic index (edge-chromatic-number)** $\chi'(G)$ of a graph G is the smallest k such that G has a proper k -edge-coloring.

Remark 5.26. $\Delta(G) \leq \chi'(G)$

Theorem 5.27 (Vizing's theorem). *Every graph G has a proper edge coloring with at most $\Delta(G) + 1$ colors. Thus $\chi'(G) \in \{\Delta(G), \Delta(G) + 1\}$.*

Proof. not here. $\square$

6 Ramsey Theory

In this chapter, if we talk about coloring, we always mean a not necessarily proper coloring (think of a child painting the graph randomly).

Example 6.1. In any group of 6 people, there are either 3 people who mutually know or don't know each other. $\iff$ Any Graph with six vertices contains a triangle or an independent set of size 3. $\iff$ Every 2-edge coloring of K_6 contains a monochromatic triangle.

Proof. Let G be a graph with 6 vertices. Pick a vertex v . v has at least 3 neighbours or non-neighbours. Wlog v has 3 neighbours a, b, c . If a, b, c are non-adjacent, we are done. Otherwise, wlog a and b are adjacent, then v, a, b is a triangle. $\square$

Definition 6.2. An **independent set** in a graph G is a set $U \subseteq V(G)$ s.t. $G[U]$ has no edges. Let $\alpha(G)$ denote the **independence number** of G , i.e. the size of the largest independent set in G .

Theorem 6.3 (Pigeonhole Principle). *Let $n_1, \dots, n_t \in \mathbb{N}$. Let X be a set with at least $1 + \sum_{i=1}^t (n_i - 1)$ elements, and let $X_1, \dots, X_t$ be disjoint sets forming a partition of X . Then there exists $i \in \{1, \dots, t\}$ s.t. $|X_i| \geq n_i$.*

Proof. Otherwise $|X| = \sum_{i=1}^t |X_i| \leq \sum_{i=1}^t (n_i - 1) < 1 + \sum_{i=1}^t (n_i - 1) \not\leq$ $\square$

Theorem 6.4 (Ramsey's Theorem). *For any $k \geq 2$, $n \in \mathbb{N}$, there exists a $N \in \mathbb{N}$ s.t. every edge coloring of K_N with k colors contains a monochromatic copy of K_n .*

Proof. Set $N = k^{kn}$. Aim: Find $v_1, \dots, v_{kn} \in V(K_N)$ s.t. for all edges from v_i to $v_{i+1}, \dots, v_{kn}$ all have same color.

We construct W and v_i inductively (v_i are the vertices we want at the end and W is a big enough set of candidates for later iteration steps).

- Set $W_0 := V(K_N)$.
- We want to show that for every $l \in \{1, \dots, nk\}$ we can find $\{v_1, \dots, v_l\}$ with $\{v_1, \dots, v_l\} \cap W_l = \emptyset$ and for every $i = 1, \dots, l$, the edges from v_i to $W_l \cup \{v_{i+1}, \dots, v_l\}$ have the same color.

To show that we always find a "next" suitable vertex in W , we look at the following:

- Take $v_{l+1} \in W_l$.

- Of the vertices in $W_l \setminus \{v_{l+1}\}$ at least $\frac{|W_l|-1}{k}$ ($|W_l| - 1$ edges and k colors) are joined to v_{l+1} by edges with the same color (pigeonhole principle 6.3).
- Take this set of vertices as the new W_{l+1} .

Now we show, that we actually have enough vertices in W to do the above construction.

Proof by induction: $|W_j| \geq k^{kn-j} - 1$.

$$|W_{l+1}| \geq \frac{|W_l|-1}{k} \geq \frac{k^{kn-l}-1-1}{k} = k^{kn-(l+1)} - 1.$$

For $i = 1, \dots, kn$, let c_i be the common color of the edges from v_i to $v_{i+1}, \dots, v_{kn}$. Since there are k colors, there is $I \subseteq \{1, \dots, kn\}$, $|I| \geq n$ s.t. $c_i = c_j \ \forall i, j \in I$. In particular, the edges in $G[\{v_i : i \in I\}]$ are monochromatic. $\square$



Theorem 6.5. Let $k, l \in \mathbb{N}$. Let G be a graph of order $\geq \binom{k+l-2}{k-1}$. Then $\omega(G) \geq k$ or $\alpha(G) \geq l$.

Proof. Like Theorem 6.4. $\square$

Definition 6.6. Let $r(k, l)$ be the minimum N s.t. every graph of order N has $\omega(G) \geq k$ or $\alpha(G) \geq l$. $r(k, l)$ is called the **Ramsey number**. Set $r(n) = r(n, n)$.

Remark 6.7. $r(3, 3) = 6$, $r(4, 4) = 18$, $43 \leq r(5, 5) \leq 48$

$r(4, 2) = 4$, $r(4, 5) = 25$

The proof of Ramsey's theorem gives $r(n) \leq 4^n$

The best known upper bound is $r(n) \leq 3.9999^n$

Proposition 6.8. $r(n) > (n-1)^2$

Proof. The task is to find a 2-edge coloring of K_N with $N = (n-1)^2$ s.t. there is no monochromatic K_n .

Split $V(K_N)$ into $n-1$ parts of size $n-1$. Color edges between different parts blue and edges within the same part red.

The red graph is a disjoint union of $n-1$ copies of K_{n-1} , so it does not contain a red K_n . Given n vertices, at least two will be in the same part, thus connected by red edge, so there cannot be a blue K_n .



$\square$

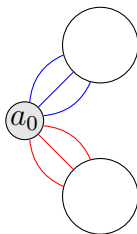
Theorem 6.9 (Erdős). $r(n) \geq 2^{\frac{n}{2}}$

Proof. Probabilistic method. Show that a random 2-edge coloring of K_N with $N = 2^{\frac{n}{2}}$ has a positive probability of containing no monochromatic K_n .

Postponed to 7 □

Theorem 6.10. *For every 2-coloring of $\binom{\mathbb{N}}{2}$, there exists an infinite set $X \subseteq \mathbb{N}$ s.t. all sets in $\binom{X}{2}$ have the same color.*

Proof. Find $a_0 \in \mathbb{N}$ and an infinite set $Y_0 \subseteq \mathbb{N} \setminus \{a_0\}$ s.t. all edges $\{a_0, y\}$ with $y \in Y_0$ have the same color. Take $a_1 \in Y_0$ and proceed analogously $\implies$ find an infinite sequence $a_0, a_1, \dots$. Because we have only colors red and blue, one of the two needs to repeat infinitely many times in the sequence. □



Remark 6.11. $\mathbb{N}$ can be replaced by any other infinite set.

7 Probabilistic Method

Definition (Rules and key ideas for this chapter). Let $n \in \mathbb{N}$, $V = \{1, \dots, n\}$, $\mathcal{G}$ is the set of all graphs with vertex set V . Aim:

- What is the probability that a random graph $G \in \mathcal{G}$ has a certain property $\mathcal{P}$?
- What is the expected value of a certain parameter of a random graph $G \in \mathcal{G}$?

Key Idea: If the probability that a random graph $G \in \mathcal{G}$ has a certain property $\mathcal{P}$ is non zero i.e. positive, then there exists a graph $G \in \mathcal{G}$ with property $\mathcal{P}$.

For every $e \in \binom{V}{2}$, draw the edge with probability $0 \leq p \leq 1$, independently. This will give us a graph G .

If G_0 is a fixed graph, then the probability that $G = G_0$ is $p^m(1-p)^{\binom{n}{2}-m}$ where $m = |E(G_0)|$.

⚠ This does not say anything about isomorphism types.

Definition 7.1. A **(discrete) probability space** is a finite set Ω (usually $\Omega = \mathcal{G}$), together with a probability function $p' : \Omega \rightarrow [0, 1]$ with the property $\sum_{\omega \in \Omega} p'(\omega) = 1$.

An **event** is a subset $A \subseteq \Omega$, the probability of A is $\mathbb{P}(A) = \sum_{\omega \in A} p'(\omega)$.

Here $\Omega = \mathcal{G}$. One needs to show that there exists a probability function $p : \mathcal{G} \rightarrow [0, 1]$ s.t. each edge occurs independently with probability p .

Let $\mathcal{G}(n, p)$ be the resulting probability space.

Remark. For countable sequences $A_1, A_2, \dots \in \Omega$, we have $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Lemma 7.2. For all $n, k \in \mathbb{N}, n \geq k \geq 2$, the probability that $G \in \mathcal{G}(n, p)$ has an independent set of size $\geq k$ is $\mathbb{P}(\alpha(G) \geq k) \leq \binom{n}{k}(1-p)^{\binom{k}{2}}$.

Proof. The probability that a fixed set $U \subseteq V$, $|U| = k$, is a independent set in G is $(1-p)^{\binom{k}{2}}$. There are $\binom{n}{k}$ such sets. By Remark 7, we have $\mathbb{P}(\alpha(G) \geq k) \leq \mathbb{P}(\bigcup_{U \subseteq V, |U|=k} \{U \text{ is an independent set}\}) \leq \sum_{U \subseteq V, |U|=k} \mathbb{P}(U \text{ is an independent set}) = \binom{n}{k}(1-p)^{\binom{k}{2}}$.

Analogously, $\mathbb{P}(\omega(G) \geq k) \leq p^{\binom{k}{2}}$. □

Proof to Theorem 6.9. Thm: $r(n) > 2^{\frac{n}{2}}$ for $n \geq 3$.

$n = 3$: $r(3) \geq 3 > 2^{\frac{3}{2}}$.

Let $n \geq 4$: We show that $\forall N \leq 2^{\frac{n}{2}}$ (where N is the order of), the probabilities for $G \in \mathcal{G}(N, \frac{1}{2})$ to have $\omega(G) \geq n$ or $\alpha(G) \geq n$ are both less than $\frac{1}{2}$.

Then $\mathbb{P}(\omega(G) \geq n \text{ or } \alpha(G) \geq n) \leq \mathbb{P}(\omega(G) \geq n) + \mathbb{P}(\alpha(G) \geq n) < \frac{1}{2} + \frac{1}{2} = 1$ so there

exists a graph G on N vertices s.t. $\omega(G) < n$ and $\alpha(G) < n$, thus $r(n) > N$.

$$\begin{aligned} \mathbb{P}(\omega(G) \geq n) &\stackrel{7.2}{\leq} \binom{N}{n} \frac{1}{2} \binom{n}{2} = \frac{N!}{(N-n)!n!} \frac{1}{2} \binom{n}{2} \stackrel{n! \geq 2^n}{\leq} \frac{N!}{(N-n)!2^n} \frac{1}{2} \binom{n}{2} \leq \frac{N^n}{2^n} \frac{1}{2} \binom{n}{2} = \frac{N^n}{2^n} \frac{1}{2} \frac{n!}{2^{(n-2)!}} = \\ &\frac{2^{\frac{n}{2}}}{2^n} 2^{\frac{-n(n-1)}{2}} = 2^{\frac{n}{2} - n - \frac{n^2}{2} + \frac{n}{2}} = 2^{-\frac{n}{2}} < 1. \end{aligned} \quad \square$$

Definition. For a probability space (Ω, p) a **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$ $[(\Omega, p) = \mathcal{G}(n, p)]$.

Graph invariants like maximal degree, girth, etc. can be interpreted as random variables. The **expectation** of X is $\mathbb{E}[X] = \sum_{\omega \in \Omega} \mathbb{P}(\omega) X(\omega)$. Note that $\mathbb{E}[x + y] = \mathbb{E}[x] + \mathbb{E}[y]$ and $\forall \lambda \in \mathbb{R} : \mathbb{E}[\lambda x] = \lambda \mathbb{E}[x]$.

Lemma 7.3. Let $k \in \mathbb{N}$, $p = p(n)$ a function s.t. $p \geq \frac{6k \ln(n)}{n}$ for large enough n . Let $G \in \mathcal{G}(n, p(n))$. Then $\lim_{n \rightarrow \infty} \mathbb{P}(\alpha(G) \geq \frac{1}{2} \frac{n}{k}) = 0$.

Proof. $\forall n, r \in \mathbb{N}$ with $n \geq r \geq 2$, we have:

$$\mathbb{P}[\alpha(G) \geq r] \stackrel{7.2}{\leq} \binom{n}{r} (1-p)^{\binom{r}{2}} \leq n^r (1-p)^{\binom{r}{2}} = (n(1-p)^{\frac{r-1}{2}})^r \stackrel{1-p \leq e^{-p}}{\leq} (ne^{-p \frac{r-1}{2}})^r.$$

Using $p \geq \frac{6k \ln(n)}{n}$ and $r \geq \frac{1}{2} \frac{n}{k}$, we have

$$ne^{-p \frac{r-1}{2}} = ne^{-\frac{pr}{2} + \frac{p}{2}} \leq ne^{-\frac{3}{2} \ln(n) + \frac{p}{2}} = n^{-1/2} e^{p/2} \leq \sqrt{\frac{e}{n}}.$$

Thus $\lim_{n \rightarrow \infty} \mathbb{P}[\alpha(G) \geq \frac{1}{2} \frac{n}{k}] \leq \lim_{n \rightarrow \infty} [(ne^{-p \frac{r-1}{2}})^r] \leq \lim_{n \rightarrow \infty} [\sqrt{\frac{e}{n}}^{\frac{n}{2k}}] = 0. \quad \square$

Lemma 7.4. Fix n, p and $k \geq 3$. Let $X : \mathcal{G}(n, p) \rightarrow \mathbb{N}$ be the number of k -cycles. The expected number of k -cycles is $\mathbb{E}[X] = \frac{(n)_k}{2k} p^k$, where $(n)_k = n(n-1)\dots(n-k+1)$.

Proof. For every possible k -cycle C with vertices in $\{1, \dots, n\}$, let $X_C : \mathcal{G}(n, p) \rightarrow \{0, 1\}$, $G \rightarrow \begin{cases} 1, & C \subseteq G \\ 0, & \text{else} \end{cases}$.

$\mathbb{E}[X_C] = \mathbb{P}(X_C = 1) = \mathbb{P}(C \subseteq G) = p^k$. There are $(n)_k$ sequences $v_1, \dots, v_k$ of distinct vertices in $\{1, \dots, n\}$, but each cycle is labeled by $2k$ such sequences so there are $\frac{(n)_k}{2k}$ possible cycles.

$$X = \sum_{C \text{ } k\text{-cycle}} X_C \text{ so } \mathbb{E}[X] = \sum_{C \text{ } k\text{-cycle}} \mathbb{E}[X_C] = \frac{(n)_k}{2k} p^k. \quad \square$$

Lemma 7.5 (Markov's inequality). Let $X : \Omega \rightarrow \mathbb{R}$. Then $\forall a > 0 : \mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$.

Theorem 5.12. Thm: For every $k \in \mathbb{N}$, there exists a graph with girth $g(G) > k$ and chromatic number $\chi(G) > k$.

We may assume $k \geq 3$.

Fix ε with $0 < \varepsilon < \frac{1}{k}$, $p = p(n) = n^{\varepsilon-1}$.

Let $X : \mathcal{G}(n, p) \rightarrow \mathbb{R}$ be a random variable assigning to every graph $G \in \mathcal{G}(n, p)$ the number of cycles of length $\leq k$.

Want to enforce $\mathbb{P}[X \geq \frac{n}{2}] < \frac{1}{2}$ (few small cycles), $\mathbb{P}[\alpha(G) \geq \frac{1}{2} \frac{n}{k}] < \frac{1}{2}$ (few independent sets).

By Lemma 7.4, $\mathbb{E}[X] = \sum_{i=3}^k \frac{(n)_i}{2^i} p^i \leq \frac{1}{2} \sum_{i=3}^k n^i p^i \leq \frac{1}{2} (k-2) n^k p^k$.

By Lemma 7.5, $\mathbb{P}[X \geq \frac{n}{2}] \leq \frac{\mathbb{E}[X]}{\frac{n}{2}} \leq (k-2) n^{k-1} p^k \stackrel{np=n^\varepsilon}{=} (k-2) n^{k-1} n^{k(\varepsilon-1)} = (k-2) n^{k-1+k\varepsilon-k} = (k-2) n^{k\varepsilon-1}$.

As $k\varepsilon - 1 < 0$, this yields $\lim_{n \rightarrow \infty} \mathbb{P}[X \geq \frac{n}{2}] = 0$. By choosing n sufficiently large, we can enforce $\mathbb{P}[X \geq \frac{n}{2}] < \frac{1}{2}$ and $\mathbb{P}[\alpha(G) \geq \frac{1}{2} \frac{n}{k}] < \frac{1}{2}$ by Lemma 7.3.

Then $\mathbb{P}[X \geq \frac{n}{2} \text{ or } \alpha(G) \geq \frac{1}{2} \frac{n}{k}] \leq \mathbb{P}[X \geq \frac{n}{2}] + \mathbb{P}[\alpha(G) \geq \frac{1}{2} \frac{n}{k}] < \frac{1}{2} + \frac{1}{2} = 1 \implies$ There exists a graph G with $< \frac{n}{2}$ cycles of length $\leq k$ and $\alpha(G) < \frac{1}{2} \frac{n}{k}$.

Now delete a vertex from each cycle of length $\leq k$, call the resulting graph H . Then $|V(H)| \geq \frac{n}{2}$, $g(H) > k$ and thus $\chi(H) \geq \frac{|V(H)|}{\alpha(H)} > \frac{\frac{n}{2}}{\frac{1}{2} \frac{n}{k}} = k$. $\square$

Definition 7.6. Fix a probability space. Events A, B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, which is equivalent to $\mathbb{P}(A|B) = \mathbb{P}(A)$ provided that $\mathbb{P}(B) \neq 0$ (where $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$).

An event is **mutually independent** from events $B_1, \dots, B_n$ if for every $I \subseteq \{1, \dots, n\}$: $\mathbb{P}(A | \bigcap_{i \in I} B_i) = \mathbb{P}(A)$.

A **dependency graph** for events $B_1, \dots, B_n$ is a graph on a vertex set $\{B_1, \dots, B_n\}$ and a set of edges s.t. $\forall i \in \{1, \dots, n\}$, the event B_i is mutually independent from $V \setminus (\{B_i\} \cup N(B_i))$.

Theorem 7.7 (Lovász Local Lemma). *Let $B_1, \dots, B_n$ be events in a (finite) probability space. If there exists $p \in \mathbb{R}$ with $0 \leq p < 1$ s.t.*

- $\mathbb{P}(B_i) < p$ for all $i \in \{1, \dots, n\}$
- the max. degree in the dependency graph for $B_1, \dots, B_n$ is at most d
- $p \leq \frac{1}{4d}$ (d is at least 1 (graph connected) so $p \leq \frac{1}{4}$)

Then $\mathbb{P}(\bigcap_{i=1}^n \overline{B_i}) > 0$ ($\overline{B}$ is the complement of B).

Proof. For $S \subseteq \{1, \dots, n\}$, we show inductively that:

- (i) $\mathbb{P}(\bigcap_{i \in S} \overline{B_i}) > 0$
- (ii) $\forall k \in \{1, \dots, n\} : \mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) \leq 2p$

Wlog assume that the dependency graph is connected.

Proof by induction on $|S|$.

$|S| = 1$, i.e. $S = \{k\}$: (i): $\mathbb{P}(\overline{B_k}) = 1 - \mathbb{P}(B_k) > 1 - p > 0$.

(ii): If B_i and B_k are mutually independent, then $\mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) = \mathbb{P}(B_k) \leq p \leq 2p$.

If B_i and B_k are not mutually independent, then $\mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) = \frac{\mathbb{P}(B_k \cap \overline{B_i})}{\mathbb{P}(\overline{B_i})} \leq \frac{\mathbb{P}(B_k)}{\mathbb{P}(\bigcap_{i \in S} \overline{B_i})} \stackrel{(i)}{\leq} \frac{p}{1-p} \leq 2p$.

Wlog $S = \{1, \dots, s\}$

(i): $\mathbb{P}(\bigcap_{i=1}^s \overline{B_i}) = \mathbb{P}(\overline{B_1})\mathbb{P}(\overline{B_2}|\overline{B_1})\dots\mathbb{P}(\overline{B_s}|\bigcap_{i=1}^{s-1} \overline{B_i}) = \prod_{i=1}^s \mathbb{P}(\overline{B_i} | \bigcap_{j=1}^{i-1} \overline{B_j}) = \prod_{i=1}^s (1 - \mathbb{P}(B_i | \bigcap_{j=1}^{i-1} \overline{B_j})) \stackrel{HI}{\geq} s(1 - 2p) > 0$.

(ii): Need to show that $\mathbb{P}(B_k | \bigcap_{i=1}^s \overline{B_i}) \leq 2p$ ($k \in \{1, \dots, n\}$).

We split $S = S_1 \cup S_2$ s.t. B_k is mutually independent from S_2 . If $S_2 = S$, then $\mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) = \mathbb{P}(B_k) < p$. So assume $S_2 \subsetneq S$.

Set $F_S = \bigcap_{i \in S} \overline{B_i} \stackrel{(1)}{\neq} \emptyset$, $F_{S_1} = \bigcap_{i \in S_1} \overline{B_i}$, $F_{S_2} = \bigcap_{i \in S_2} \overline{B_i}$.

We want to show that $\mathbb{P}(B_k | F_S) \leq 2p$.

We have: $\mathbb{P}(B_k | F_S) = \frac{\mathbb{P}(B_k \cap F_S)}{\mathbb{P}(F_S)} = \frac{\mathbb{P}(B_k \cap F_{S_1} | F_{S_2}) \mathbb{P}(F_{S_2})}{\mathbb{P}(F_{S_1} | F_{S_2}) \mathbb{P}(F_{S_2})} = \frac{\mathbb{P}(B_k \cap F_{S_1} | F_{S_2})}{\mathbb{P}(F_{S_1} | F_{S_2})} = (*)$

We have: $\mathbb{P}(B_k \cap F_{S_1} | F_{S_2}) \leq \mathbb{P}(B_k | F_{S_2}) \leq \mathbb{P}(B_k) \leq p$.

For the denominator, we calculate: $\mathbb{P}(F_{S_1} | F_{S_2}) = 1 - \mathbb{P}(\bigcup_{i \in S_1} B_i | F_{S_2}) \geq 1 - \sum_{i \in S_1} \mathbb{P}(B_i | F_{S_2}) \stackrel{IH}{\geq}$

$1 - \sum_{i \in S_1} 2p \stackrel{S_2 \subsetneq S}{\geq} 1 - 2pd \stackrel{p \leq \frac{1}{4d}}{\geq} \frac{1}{2}$.

$\implies (*) \leq \frac{p}{\frac{1}{2}} = 2p$. □

Remark 7.8. • The bounds can be slightly improved

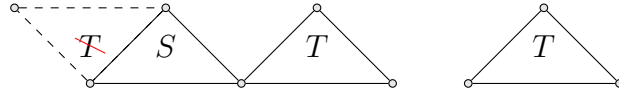
- Asymmetric version exists
- There is an algorithmic version, the Moser-Tardos-Algorithm

Lemma 7.9. Let $k, n \in \mathbb{N}_{\geq 2}$ s.t. $4 \binom{k}{2} \binom{n-k}{k-2} 2^{1-\binom{k}{2}} < 1$.

Then there exists a 2-edge coloring of K_n that does not contain a monochromatic K_k . Thus $r(k) > n$.

Proof. Consider a random 2-edge coloring of K_n . For $S \subseteq \{1, \dots, n\}$ of size k , consider the event A_S that the graph induced by S is monochromatic. Thus $\mathbb{P}(A_S) = \frac{2}{2^{\binom{k}{2}}} = 2^{1-\binom{k}{2}}$. Set $p = 2^{1-\binom{k}{2}} + \varepsilon$ with $\varepsilon > 0$.

A_S is mutually independent from the events A_T with $|S \cap T| \leq 1$.



Thus the maximal degree of the corresponding dependency graph satisfies $d \leq \binom{k}{2} \binom{n-2}{k-2}$.

Then $p \leq \frac{1}{4d}$. Thus by Theorem 7.7 we have $\mathbb{P}\left(\bigcap_{S \subseteq \{1, \dots, n\}, |S|=k} \overline{A_S}\right) > 0$ ($\mathbb{P}(\text{no monochromatic } K_k) > 0$).

Thus there exists a 2-edge coloring of K_n that does not contain a monochromatic K_k . $\square$

Remark 7.10. This yields $r(k) > \frac{\sqrt{2}}{4}(1 + o(1))k2^{\frac{k}{2}}$.

8 Tree Decompositions And Treewidth

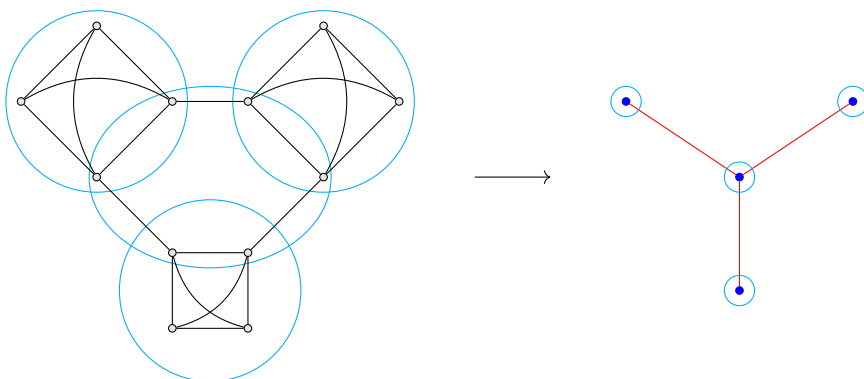
Definition 8.1. A **tree decomposition** of a graph $G = (V, E)$ is a pair $D = (S, T)$, where $S = \{X_i : i \in I\}$ is a set of subsets of V and $T = (I, F)$ is a tree such that the following hold:

- (i) $\bigcup_{i \in I} X_i = V$;
- (ii) For every edge $\{u, v\} \in E$, there is an $i \in I$ such that $u, v \in X_i$;
- (iii) For all $v \in V$, the subtree of T induced by $\{i \in I : v \in X_i\}$ is connected;

The sets X_i are called **bags**.

The **width** of D is $wd(D) = \max_{i \in I} |X_i| - 1$.

The **treewidth** of G is $tw(G) = \min\{wd(D) : D \text{ is a tree decomposition of } G\}$.



Remark 8.2. Every graph has a trivial tree decomposition of width $|V(G)| - 1$, which has a single bag containing all vertices of G .

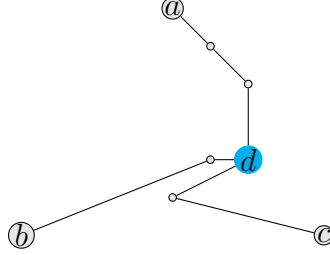
Remark 8.3. Equivalent formulations for (iii):

- (iii') If, for $v \in V$, we have $v \in X_i$ and $v \in X_j$, then $v \in X_k$ for all $k \in I$ that lie on the unique path from i to j in T .
- (iii'') For $i, j \in I$ and $k \in I$ on the unique path from i to j , $X_i \cap X_j \subseteq X_k$.

Since the treewidth of a graph G is the maximum of the treewidths of its connected components, we will focus on connected graphs. We have $G = K_1 \iff tw(G) = 0$, we assume $|V(G)| \geq 2$ from now on.

Lemma 8.4. Let G be a connected graph with tree decomposition $D = (S, T)$. For every clique $C \subseteq V(G)$, there exists a bag X_i , with $C \subseteq X_i$.

Proof. Induction on $k = |C|$. For $k = 1$, this is (i), for $K = 2$ it's (ii) (Definition 8.1). Now let $k \geq 3$ and let u, v, w be distinct vertices in C . By IH, there exist bags X_a, X_b, X_c s.t. $C \setminus \{u\} \subseteq X_a$, $C \setminus \{v\} \subseteq X_b$, $C \setminus \{w\} \subseteq X_c$. If any two of a, b, c are equal, the corresponding bag contains C . Thus let a, b, c be pairwise distinct. Let P_{ab}, P_{ac}, P_{bc} be the unique paths between ab, ac, bc in T . They must share a vertex $d \notin \{a, b, c\}$ since concatenating P_{ab}, P_{ac}, P_{bc} would be a cycle otherwise, but T is a tree.



By (iii'') from 8.3, we have $C \setminus \{u, v\} \subseteq X_a \cap X_b \subseteq X_d$. Similarly, $C \setminus \{u, w\} \subseteq X_d$ and $C \setminus \{v, w\} \subseteq X_d \implies C \subseteq X_d$. $\square$

Corollary 8.5. *For every graph G , we have $\omega(G) - 1 \leq tw(G)$.*

Lemma 8.6. *If H is a minor of G , then $tw(H) \leq tw(G)$*

Proof. If H is obtained from G by deleting a vertex or an edge, then $tw(H) \leq tw(G)$ as every tree decomposition of G remains one for H (after deleting the vertex/vertices from all bags containing them).

Suppose H is obtained from G by contracting an edge $\{u, v\}$.

Let $D = (S, T)$ be a tree decomposition of G . Replace all occurrences of u and v in the bags by a new vertex w . Since bags containing u induce a subtree of T , similarly for v , and there is a bag containing u and v , the bag containing w induces a subtree in T as well. $\square$

Lemma 8.7. *Let $G = (V, E)$ connected with $|V| \geq 2$. Then $tw(G) = 1 \iff G$ is a tree.*

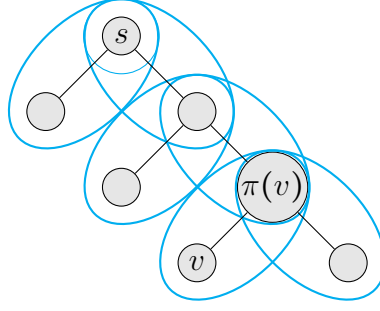
Proof. " $\Leftarrow$ "

Let G be a tree. Choose $s \in V$ and run DFS (Depth-First Search) on G . This yields for every $v \in V \setminus \{s\}$ a unique predecessor $\pi(v)$.

Set $X_s = \{s\}$, $X_v = \{v, \pi(v)\} \forall v \in V \setminus \{s\}$, $T = G$, $S = \{X_v : v \in V\}$.

We claim that $D = (T, S)$ is a tree decomposition of G .

Properties (i) and (ii) directly follow from the constructions. For $v \in V$, the set of bags containing v is X_v together with all X_w where w is a child of v . Thus the subtree induced by the corresponding vertices in T is connected. $\implies tw(G) = 1$ (bags only have a max. of 2 vertices in them).



" $\implies$ "

If G is not a tree $\implies G$ contains a cycle and hence a K_3 minor. By Corollary 8.5, we have $tw(K_3) = 2$ and by Lemma 8.6, $2 = tw(K_3) \leq tw(G) \not\leq 1 \implies G$ is a tree. $\square$

Remark. Computing treewidth is NP-hard

Theorem 8.8 (Bodlaender). *There exists an algorithm that, given a graph G with $tw(G) = k$, computes a tree decomposition of G of width k in time: $2^{\mathcal{O}(k^3)}|V(G)|$*

Remark. • Bodlaender et al. (2016): An algorithm that constructs a treedecomposition of width $\leq 5k + 4$ or decides $tw(G) \geq k$ in $\mathcal{O}(2^{\mathcal{O}(k)}n)$

- Korhonen (2021): width $\leq 2k + 1$ with same runtime
- Korhonen, Lokshtanov (2023): width k , runtime $2^{\mathcal{O}(k^2)}\mathcal{O}(n^4)$
- open question: What is the complexity of computing treewidth on planar graphs. Is it in P?

Definition 8.9. A **nice treedecomposition** $D = (S, T)$ of a graph $G = (V, E)$ is a tree decomposition in which a root vertex can be chosen s.t. T is a binary tree (i.e. every vertex has at most 2 children) and such that every vertex of T is of one of the following types:

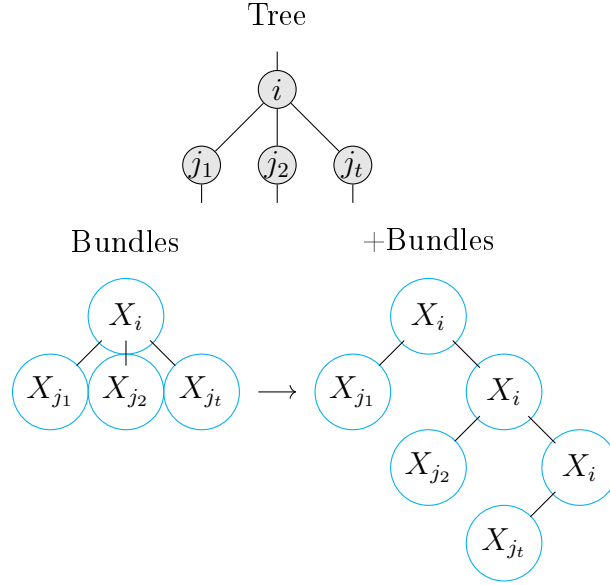
- **leaf**: if $i \in V(T)$ is a leaf of T , then $|X_i| = 1$
- **introduce node**: if $i \in V(T)$ has exactly one child j and $X_i = X_j \cup \{v\}$ for some $v \in V(G)$
- **forget node**: if $i \in V(T)$ has exactly one child j and $X_i = X_j \setminus \{v\}$ for some $v \in X_j$
- **join node**: if $i \in V(T)$ has precisely two children j_1, j_2 and $X_i = X_{j_1} = X_{j_2}$

Theorem 8.10. *Let $D = (S, T)$ a tree decomposition of $G = (V, E)$. Then we can construct a nice treedecomposition $D' = (S', T')$ of G with $tw(D') \leq tw(D)$, $|V(T')| \leq ctw(D)|V(T)|$ in polynomial time.*

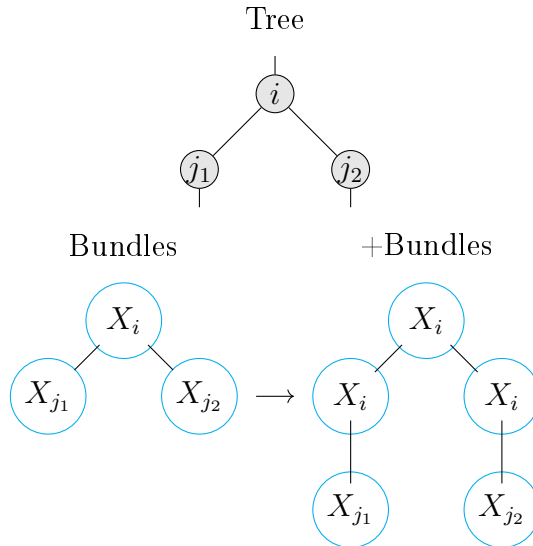
Proof. Root T at an arbitrary vertex $s \in V(T)$.

(i) Make T binary:

For every $i \in V(T)$ that has ≥ 3 children $j_1, \dots, j_t$, replace i and its descendants as follows:



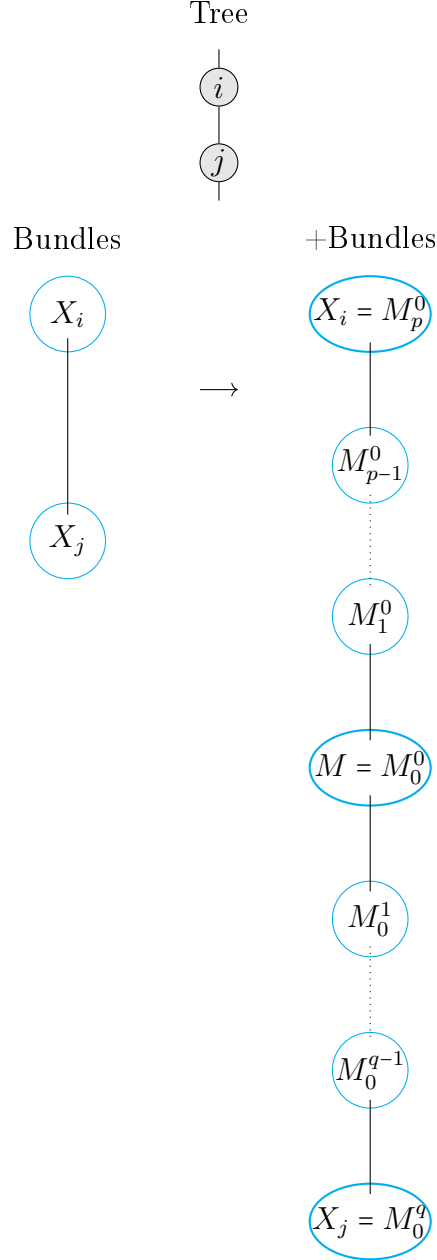
(ii) Make all vertices with precisely two children join nodes:



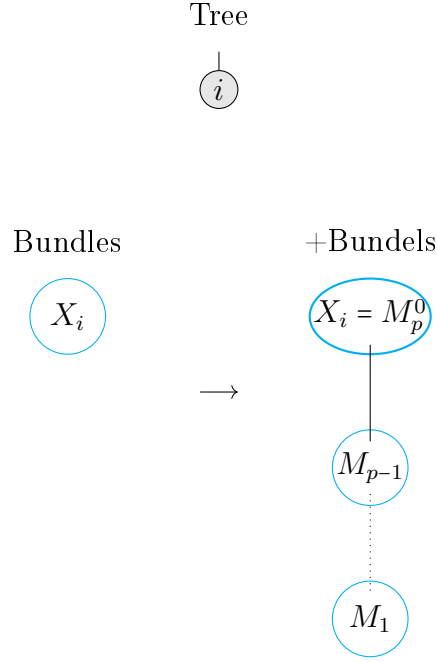
(iii) Make vertices with precisely one child introduce/forget nodes:

Let $i \in V(T)$ with a single child j . Set $M = X_i \cap X_j$ and write $X_i =$

$M \dot{\cup} \{v_1, \dots, v_p\}$, $X_j = M \dot{\cup} \{u_1, \dots, u_q\}$ with $\{v_1, \dots, v_p\} \cap \{u_1, \dots, u_q\} = \emptyset$. Now add bags $M_a^b = M \dot{\cup} \{v_1, \dots, v_a\} \cup \{u_1, \dots, u_b\}$. Replace the edge $\{i, j\}$ by M_a^b for $a \in \{0, \dots, p\}, b = 0$ and $b \in \{1, \dots, q\}, a = 0$. Connect M_a^b with M_{a-1}^b if $a \neq 0$ and to M_a^{b+1} if $a = 0$.



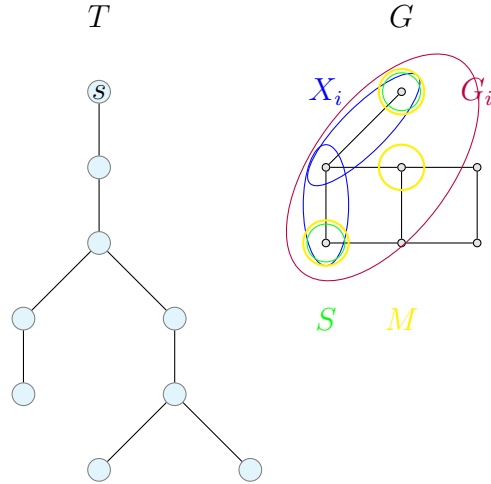
(iv) Leafs: Let $i \in V(T)$ be a leaf. Write $X_i = \{v_1, \dots, v_p\} = M_p$



This yields a nice tree decomposition D' with $tw(D)|V(T)|c$ vertices and $tw(D') \leq tw(D)$. $\square$

Lemma 8.11. *Let $G = (V, E)$ be a graph and let $D = (S, T)$ be a nice treedecomposition of $tw(D) = k$. An independent set in G of size $\alpha(G)$ can be computed in $\mathcal{O}(k2^{k+1}n)$, where $n = |V(G)|$.*

Proof. For $i \in V(T)$, let $V_i = \textcolor{blue}{X}_i \cup \bigcup_{j \text{ descendent of } i \text{ in } T} X_j$. Set $\textcolor{red}{G}_i = G[V_i]$.



For all $i \in V(T)$ and $S \subseteq X_i$, let $C_i(S) = \max\{|M| : M \text{ independent set in } G_i, M \cap X_i = S\}$. Set $C_i(S) = -\infty$ if the above set is empty.

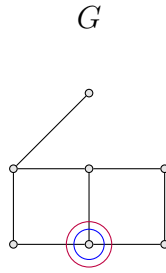
For every $i \in V(T)$ make a table C_i containing $C_i(S) \quad \forall S \subseteq X_i$. Then C_i has $2^{|X_i|} \leq 2^{k+1}$ entries.

If we manage to compute $C_i(S)$ for all i and S then $\alpha(G) = \max\{|M| : \text{independent set in } G\} = \max\{C_s(S) : S \subseteq X_s\}$ with s root of T .

Note $G_s = G$. This can be computed in $\mathcal{O}(2^{k+1})$.

Now compute $C_i(S)$ starting from leafs of T .

- (i) $i \in V(T)$ leaf: $X_i = \{v\}$, $G_i = G[V_i] = G[\{v\}] = \{v\}$. $S = \emptyset \implies C_i(\emptyset) = 0$,
 $S = \{v\} \implies C_i(S) = C_i(\{v\}) = 1$.



- (ii) $i \in V(T)$ introduce node: Let j be the unique child of i in T , $X_i = X_j \cup \{v\}$ for some $v \in V(G) \setminus X_j$.

Claim: v is not adjacent to any vertex of $V_i \setminus X_j$.

Let $w \in V_i \setminus X_j$. Then by property (iii) 8.1, all vertices of T corresponding to bags are descendants of j . By property (iii) 8.1, since $v \notin X_j$, v is not contained in any bag corresponding to a descendant of j in T . Thus v and w are not contained in a joint bag, so by property (ii) 8.1 v and w are non-adjacent.

Let $S \subseteq X_i = X_j \cup \{v\}$. If $v \notin S$, then $C_i(S) = C_j(S)$.

If $v \in S$, then $C_i(S) = \begin{cases} -\infty & \text{if } \{v, w\} \in E \text{ for some } w \in S \\ C_j(S \setminus \{v\}) + 1 & \text{if } \{v, w\} \notin E \text{ for all } w \in S \end{cases}$ since

v is not adjacent to any vertex of $V_j \setminus X_j$ by Property (iii) 8.1.

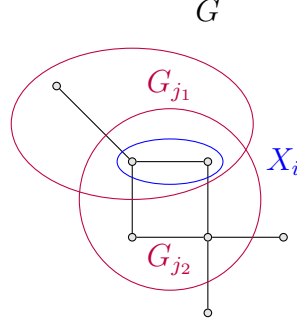
(take independent set $M' \subseteq V_j$ with $M' \cap X_j = S \setminus \{v\}$ and $|M'| = C_j(S \setminus \{v\})$. By claim no vertex in $M' \setminus X_j$ is adjacent to v . Thus if $\{v, w\} \notin E$ for all $w \in S$, then $M' \cup \{v\}$ is independent so $C_i(S) \geq C_j(S \setminus \{v\}) + 1$, and that is optimal.) Thus, if we know C_j , we can compute C_i in $\mathcal{O}(k2^{k+1})$.

- (iii) $i \in V(T)$ forget node: Let j be the unique child of i in T , $X_i = X_j \setminus \{v\}$ for some $x \in X_j$. Then $V_i = V_j$, $G_i = G_j$. For $S \subseteq X_i$, we have $C_i(S) = \max\{C_j(S), C_j(S \cup \{v\})\}$. This is computable in $\mathcal{O}(2^{k+1})$.

(iv) $i \in V(T)$ join node:

Let j_1, j_2 be the two children of i in T . Then $X_i = X_{j_1} = X_{j_2}$.

CLaim: $\forall S \subseteq X_i$, $C_i(S) = C_{j_1}(S) + C_{j_2}(S) - |S|$ (so $C_i(S)$ is computable in $O(1)$).



Proof of claim:

" $\leq$ ": M is independent set in G_i with $M \cap X_i = S$. Then $M \cap V_{j_1}$ is independent in G_{j_1} and $M \cap V_{j_2}$ is independent in G_{j_2} .

$$M \cap X_{j_1} = M \cap X_i = S = M \cap X_{j_2}$$

$$C_i(S) = |M| = |M \cap V_{j_1}| + |M \cap V_{j_2}| - |M \cap V_{j_1} \cap V_{j_2}| \leq C_{j_1}(S) + C_{j_2}(S) - |S|.$$

" $\geq$ ": Let M_1 and M_2 be independent sets in G_{j_1} and G_{j_2} with $M_1 \cap X_{j_1} = S = M_2 \cap X_{j_2}$, $|M_l| = C_{j_l}(S)$ for $l = 1, 2$.

We show that $M = M_1 \cup M_2$ is independent in G_i .

Let $v, w \in M$ and assume $\{v, w\} \in E(G_i)$. Then wlog $v \in V_{j_1}$, $w \in V_{j_2}$ (M_1, M_2 are independent). If both $v \notin X_i$ and $w \notin X_i$, then $\{v, w\} \notin E(G_i)$ by property (ii) 8.1 $\nmid$.

So wlog $v \in X_i$. Then $v \in X_i \cap M_1 = X_{j_1} \cap M_1 = S \subseteq M_2$, $w \in M_2 \nmid$ because M_2 is independent.

Thus $M = M_1 \cup M_2$ is independent. So $C_i(S) \geq |M| = |M_1| + |M_2| - |M_1 \cap M_2| = C_{j_1}(S) + C_{j_2}(S) - |S|$

Thus we can compute $C_i(S)$ for all S in $O(k2^{k+1}n)$. $\alpha(G) = \max\{C_s(S) : S \subseteq X_s\}$ where s is the root of T . This maximum can be computed in $O(2^{k+1})$ so $O(k2^{k+1}n)$ in total. $\square$

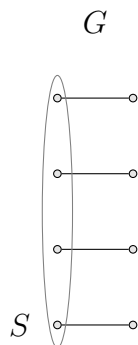
Remark 8.12. • $tw(G) \leq k \implies |E| \leq k|V(G)| - \frac{1}{2}k(k+1)$

- Courcelle's theorem states that every graph problem expressible in MSO (Monadic Second Order Logic) can be solved in linear time on a graphclass of bounded treewidth.

9 Expander Graphs

Aim: Sparse graphs with "strong connectivity" properties.

Definition 9.1 (Edge expansion). Let $G = (V, E)$ be a graph, $|V| = n$. For $S \subsetneq V$, let ∂S be the set of edges with exactly one endpoint in S . The **edge expansion** (isoperimetric number or cheeger constant) of G is $h(G) = \min_{S \subsetneq V, 0 < |S| \leq \frac{n}{2}} \frac{|\partial S|}{|S|}$.



Remark 9.2. For every connected graph G with $|V| = n$, $h(G) > 0$.

Definition 9.3. Let $\varepsilon > 0$. A sequence of (d -regular) graphs $(G_i)_{i \in \mathbb{N}}$ of order increasing with i is called an ε -**sequence of expanders** (or ε -expander sequence) if $h(G_i) > \varepsilon \quad \forall i \geq 1$. If there exists any $\varepsilon > 0$ with this property, we call (G_i) an expander sequence.

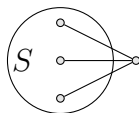
Remark 9.4. Without the assumption on the max. degree, $K_1, K_2, K_3, \dots$ is an expander sequence.

Definition 9.5 (Vertex expansion). Let $G = (V, E)$ be a graph. For $S \subseteq V$, let $N(S) = \{v \in V \setminus S : v \in N(u) \text{ for some } u \in S\}$.

Then $h_{out}(G) = \min_{S \subseteq V, 0 < |S| \leq \frac{n}{2}} \frac{|N(S)|}{|S|}$ is the **vertex expansion** of G .

Lemma 9.6. Let $G = (V, E)$ be a d -regular graph. Then $h_{out}(G) \leq h(G) \leq d \cdot h_{out}(G)$.

Proof. For $S \subseteq V$, $0 < |S| \leq \frac{n}{2}$, we have $|N(S)| \leq |\partial S| \leq d \cdot |N(S)|$ (every vertex is incident to at least 1 and at most d edges).



□

Example 9.7. • (Margulis 1975; Gabber-Galil 1981): Let G_k be a graph with vertex set $\mathbb{Z}_k \times \mathbb{Z}_k$. A vertex (x, y) is adjacent to $(x \pm 2y, y), (x \pm (2y + 1), y), (x, y \pm 2x), (x, y \pm (2x + 1))$.

Thus G_k is a sequence of 8-regular graphs.

This is a sequence of expanders. The proof relies on harmonic analysis and representation theory.

- Let G_p be a graph with vertex set $\mathbb{Z}_p$ (p prime) and $x \in \mathbb{Z}_p$ is adjacent to $x \pm 1$ and x^{-1} ($0^{-1} = 0$).

This is not a simple graph!!! Proof of expansion relies on number theory.

- zigzag product (Reinhold, Vadham, Wigderson 2002): Given two graphs G, H produces a graph with only slightly worse expansion properties.
- Many randomized constructions

9.1 Spectral Theory for Expanders

Let G be a d -regular connected graph on n vertices. Let $A(G)$ be its adjacency matrix. By the spectral theorem $A(G)$ has real eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. The multiset of the eigenvalues of $A(G)$ is called the **spectrum of G** .

One can show that:

- $\lambda_1 = d$
- $\lambda_1, \dots, \lambda_n \in [-d, d]$
- $\lambda_n = -d \iff G$ is bipartite
- G connected $\iff \lambda_1 > \lambda_2$

Lemma 9.8 (Cheeger inequality). *Let G be a d -regular (connected) graph with spectrum $\lambda_1 > \lambda_2 \geq \dots \geq \lambda_n$. Then $\frac{d-\lambda_2}{2} \leq h(G) \leq \sqrt{2d(d-\lambda_2)}$ (Note that $d-\lambda_2 = \lambda_1 - \lambda_2 > 0$).*

Definition 9.9. In this case, $d - \lambda_2$ is called the **spectral gap** of G .

Theorem 9.10 (Expander Mixing Lemma). *Let G be a d -regular connected graph on n vertices.*

Set $\lambda = \lambda(G) = \max\{|\lambda_2|, |\lambda_n|\}$.

For $S, T \subseteq V(G)$, let $e(S, T) = \{(x, y) \in S \times T : (x, y) \in E(G)\}$ then $|e(S, T) - \frac{d|S||T|}{n}| \leq \lambda \sqrt{|S||T|}$ (where the fraction is the expected number of edges between S and T in a random d -regular graph).

Proof. Let $v_1, \dots, v_n$ be eigenvectors to $\lambda_1, \dots, \lambda_n$ s.t. $\{v_1, \dots, v_n\}$ is an ONB of $\mathbb{R}^n$.

Choose $v_1 = \frac{1}{\sqrt{n}}(1, \dots, 1)^T$ and let $J = \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix} \in \mathbb{R}^{n \times n}$.

Let 1_S and 1_T be the vectors in $\mathbb{R}^n$ that have a 1 if the corresponding vertex is in S or T respectively and 0 otherwise.

Then $|e(S, T) - \frac{d|S||T|}{n}| = |1_S^T(A(G) - \frac{d}{n}J)1_T|$ ($A(G) - \frac{d}{n}J =: M \in \mathbb{R}^{n \times n}$).

Note that $Jv_1 = nv_1$ and since $v_2, \dots, v_n$ are orthogonal to v_1 , we have $Jv_i = 0$ for $i = 2, \dots, n$.

Thus M has eigenvalues $0, \lambda_2, \dots, \lambda_n$ with eigenvectors $v_1, \dots, v_n$.

By Cauchy-Schwarz, we have $|1_S^T M 1_T| \leq \|1_S^T\| \|M 1_T\|$.

$$\begin{aligned} \|M 1_T\|^2 &= \langle M 1_T, M 1_T \rangle \stackrel{M \text{ self-adjoint}}{=} \langle 1_T, M^2 1_T \rangle = \langle 1_T, \sum_{i=1}^n M^2 \langle 1_T, v_i \rangle v_i \rangle = \langle 1_T, \sum_{i=1}^n \langle 1_T, v_i \rangle M^2 v_i \rangle = \\ &= \langle 1_T, \sum_{i=2}^n \langle 1_T, v_i \rangle \lambda_i^2 v_i \rangle = \sum_{i=2}^n \lambda_i^2 \langle 1_T, v_i \rangle^2 \leq \lambda^2 \|1_T\|^2 \quad (\text{with } M^2 v_i = \begin{cases} 0 & i = 1 \\ \lambda_i^2 v_i & \text{else} \end{cases}). \end{aligned}$$

Hence $|e(S, T) - \frac{d|S||T|}{n}| = |1_S^T M 1_T| \leq \|1_S\| \|M 1_T\| \leq \lambda \|1_S\| \|1_T\| = \lambda \sqrt{|S||T|}$. $\square$

Corollary 9.11. *Let G be a d -regular graph on n vertices. Then $\alpha(G) \leq \frac{\lambda}{d}n$ and $\chi(G) \geq \frac{d}{\lambda}$.*

Proof. If S is independent, then $e(S, S) = 0$. Hence by 9.10 $\frac{d|S|^2}{n} \leq \lambda|S| \implies |S| \leq \frac{\lambda n}{d}$.

Every color class of a coloring of G is an independent set. Hence $\frac{n}{\chi(G)} \leq \alpha(G) \implies \chi(G) \geq \frac{d}{\lambda}$. $\square$

Remark 9.12. One can find bounds on the diameter of G in terms of λ . Diameter = maximal length of a shortest path between two vertices.

Theorem 9.13 (Friedman 2003). *For every $\varepsilon > 0$ and a random d -regular graph on n vertices, $\mathbb{P}(\lambda(G) \leq 2\sqrt{d-1} + \varepsilon) = 1 - o_n(1)$.*

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